



AthenaStudies

Logic & Sets Day 1

Midterm crash course | Mara Spadon | 2023

Welcome!



Mara Spadon

About me

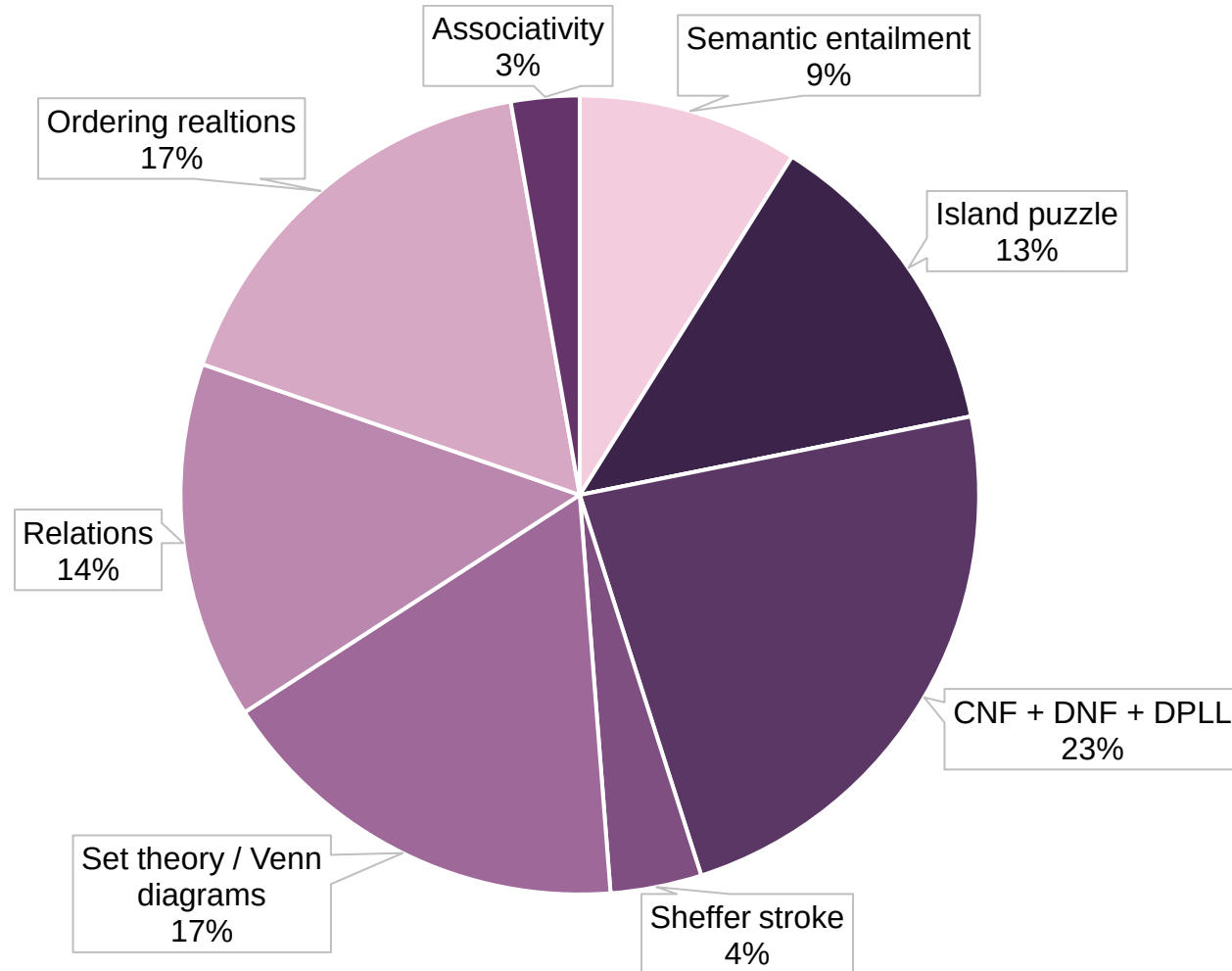
- Third year AI student
- Experience with teaching for Athena
- Hobbies: music and languages!

WhatsApp



Exam Analysis

Midterms 2019-2022

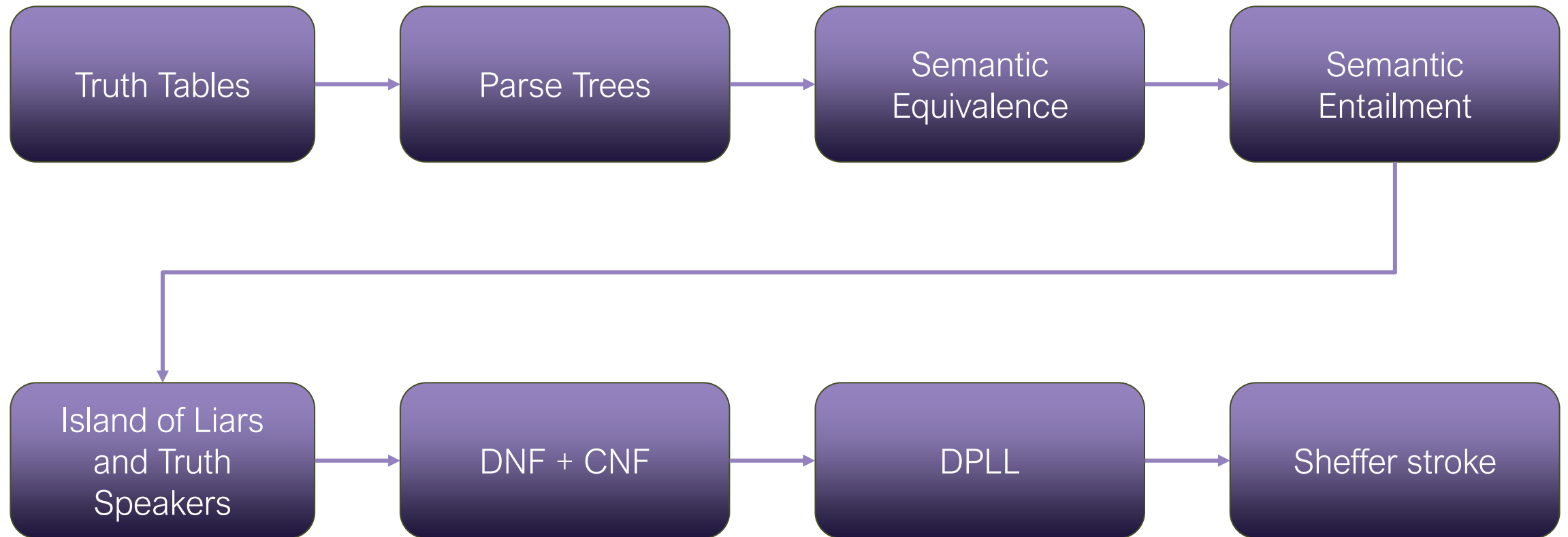


Planning

10-15 min break
every 45 min!

Sunday	Logic week 1: ○ Truth tables ○ Parse trees ○ Semantic equivalence Logic week 2: ○ Semantic entailment ○ Island of liars and truth speakers Logic week 3: ○ DNF + CNF ○ DPLL ○ Sheffer stroke
Tuesday	Sets week 1 + 2 + 3

Planning Day 1



Pro-active participation

- Camera's on (optional)
- Raise your hand to ask questions!
- Be interactive!
- Ask questions in the chat!
- Take notes!
- It is okay to make mistakes
- Slides will be sent afterwards



Structure



Structure based on past exams



Learn the most by asking questions



Active learning with old exam questions is the basis of our trainings



24/7 WhatsApp aid

Truth Tables

Propositional Formulas

The propositional logic language is made up of:

- Propositional variables
- Connectives

Propositional variables:

$p, q, r, \dots$

Connectives:

$\wedge$	'and'	<i>conjunction</i>
$\vee$	'or'	<i>disjunction</i>
$\oplus$	'either ... or ...'	<i>exclusive or</i>
$\neg$	'not'	<i>negation</i>
$\rightarrow$	'if ... then ...'	<i>implication</i>
$\leftrightarrow$	'if and only if'	<i>bi-implication</i>

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$\rightarrow$	'if ... then ...'	<i>implication</i>
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Together they form formulas!

- $((p \wedge q) \vee \neg r) \rightarrow p$
- $(p \vee q)$
- r

Truth Tables

- A formula (ϕ, ψ, χ) can be either true or false $\rightarrow$ truth value

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$\wedge$: true if both variables
are true

ϕ	ψ	$\phi \wedge \psi$
T	T	T
T	F	
F	T	
F	F	

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T	T	T
T	F	F
F	T	F
F	F	F

$\vee$: true if one or both variables is true

ϕ	ψ	$\phi \vee \psi$
T	T	
T	F	
F	T	
F	F	

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F	T	F
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$\vee$: true if one or both variables is true

ϕ	ψ	$\phi \vee \psi$
T	T	T
T	F	T
F	T	T
F	F	F

$\oplus$: true if one variable is true, not both

ϕ	ψ	$\phi \oplus \psi$
T	T	
T	F	
F	T	
F	F	

$\leftrightarrow$
T, T
F, F

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$\rightarrow$: false if ϕ is true and ψ is false. True otherwise

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Truth Tables

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T	T	T
T	F	F
F	T	T
F	F	T

$\leftrightarrow$: true if ϕ and ψ have the same value

ϕ	ψ	$\phi \leftrightarrow \psi$
T	T	
T	F	
F	T	
F	F	

Truth Tables

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F	T	F
F	F	T

$\neg$: true if ϕ is false

ϕ	$\neg \phi$
T	F
F	T

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T	T	T
T	F	F
F	T	F
F	F	T

$\neg$: true if ϕ is false

ϕ	$\neg \phi$
T	F
F	T

Truth Tables

- **Tautology**: formula is always true $\rightarrow$ all rows in truth table true
- **Contradiction**: formula is always false $\rightarrow$ all rows in truth table false
- **Contingent**: formula can be true or false $\rightarrow$ rows in truth table contain both true and false

p	$\neg p$	$p \vee \neg p$	$p \wedge \neg p$
T	F	T	F
F	T	T	F



Contingency Tautology Contradiction

Truth Tables: Example

Construct the truth table of the following formula:

$$(p \vee q) \wedge (\neg r \rightarrow q)$$

Truth Tables: Example

Construct the truth table of the following formula:

$$(p \vee q) \wedge (\neg r \rightarrow q)$$

p	q	r	$p \vee q$	$\neg r$	$\neg r \rightarrow q$	$(p \vee q) \wedge (\neg r \rightarrow q)$
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

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T	T	T	T			
T	T	F	T			
T	F	T	T			
T	F	F	T			
F	T	T	T			
F	T	F	T			
F	F	T	F			
F	F	F	F			

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p	q	r	$p \vee q$	$\neg r$	$\neg r \rightarrow q$	$(p \vee q) \wedge (\neg r \rightarrow q)$
T	T	T	T	F		
T	T	F	T	T		
T	F	T	T	F		
T	F	F	T	T		
F	T	T	T	F		
F	T	F	T	T		
F	F	T	F	F		
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Construct the truth table of the following formula:

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p	q	r	$p \vee q$	$\neg r$	$\neg r \rightarrow q$	$(p \vee q) \wedge (\neg r \rightarrow q)$
T	T	T	T	F	T	
T	T	F	T	T	T	
T	F	T	T	F	T	
T	<u>F</u>	F	T	<u>T</u>	<u>F</u>	
F	T	T	T	F	T	
F	T	F	T	T	T	
F	F	T	F	F	T	
F	<u>F</u>	F	F	<u>T</u>	<u>F</u>	

Truth Tables: Example

Construct the truth table of the following formula:

$$(p \vee q) \wedge (\neg r \rightarrow q)$$

p	q	r	$p \vee q$	$\neg r$	$\neg r \rightarrow q$	$(p \vee q) \wedge (\neg r \rightarrow q)$
T	T	T	T	F	T	T
T	T	F	T	T	T	T
T	F	T	T	F	T	T
T	F	F	T	T	F	F
F	T	T	T	F	T	T
F	T	F	T	T	T	T
F	F	T	F	F	T	F
F	F	F	F	T	F	F

Truth Tables: Exercise!

For the following formulas, compute the complete truth table, and indicate whether it is a tautology, a contingent formula, or a contradiction:

(a) $((p \rightarrow \neg q) \rightarrow \neg p) \rightarrow q$

(b) $(p \rightarrow q) \vee (p \rightarrow \neg q)$

Truth Tables: Exercise!

For the following formulas, compute the complete truth table, and indicate whether it is a tautology, a contingent formula, or a contradiction:

(a) $((p \rightarrow \neg q) \rightarrow \neg p) \rightarrow q$

(b) $(p \rightarrow q) \vee (p \rightarrow \neg q)$

a)

p	q	$\neg q$	$\neg p$	$p \rightarrow \neg q$	$(p \rightarrow \neg q) \rightarrow \neg p$	$\phi \rightarrow q$
T	T	F	F	F	T	T
T	F	T	F	T	F	T
F	T	F	T	T	T	T
F	F	T	T	T	T	F

↪ contingent formula

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a)

p	q	$\neg p$	$\neg q$	$p \rightarrow \neg q$	$(p \rightarrow \neg q) \rightarrow \neg p$	$((p \rightarrow \neg q) \rightarrow \neg p) \rightarrow q$
T	T	F	F	F	T	T
T	F	F	T	T	F	T
F	T	T	F	T	T	T
F	F	T	T	T	T	F

Both true and false valuations, so **contingent** formula

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(a) $((p \rightarrow \neg q) \rightarrow \neg p) \rightarrow q$

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b)

p	q	$\neg q$	$p \rightarrow q$	$p \rightarrow \neg q$	$\phi \vee \psi$
T	T	F	T	F	T
T	F	T	F	T	T
F	T	F	T	T	T
F	F	T	T	T	T

$\Rightarrow$ tautology

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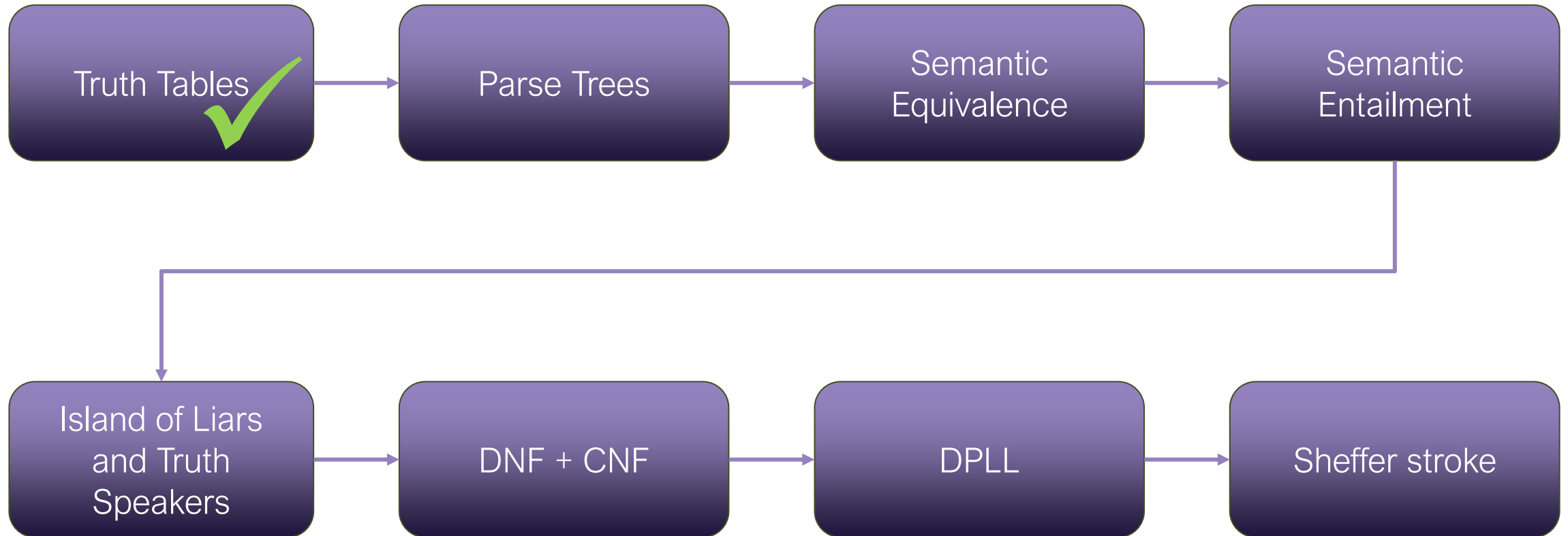
(b) $(p \rightarrow q) \vee (p \rightarrow \neg q)$

b)

p	q	$\neg q$	$p \rightarrow q$	$p \rightarrow \neg q$	$(p \rightarrow q) \vee (p \rightarrow \neg q)$
T	T	F	T	F	T
T	F	T	F	T	T
F	T	F	T	T	T
F	F	T	T	T	T

Only true valuations, so **tautology**

Planning Day 1



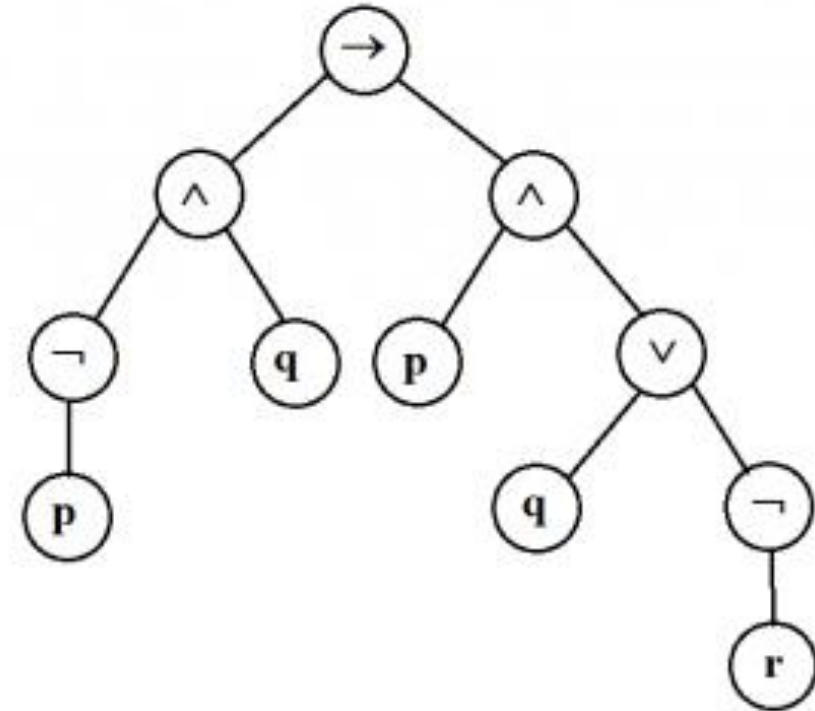
Parse Trees

Parse Trees

- Parse trees are a way to write down formulas

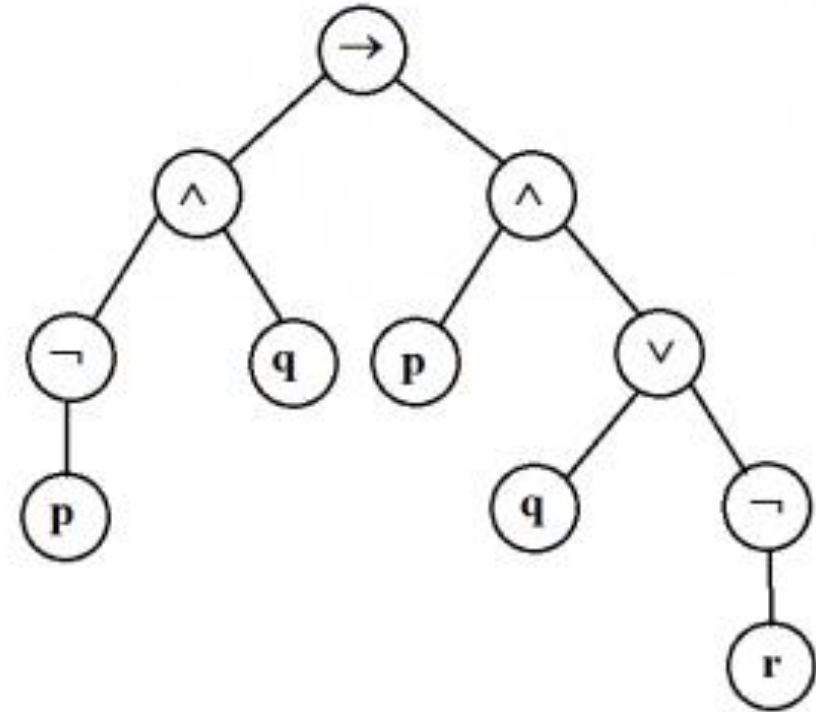
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Parse Trees

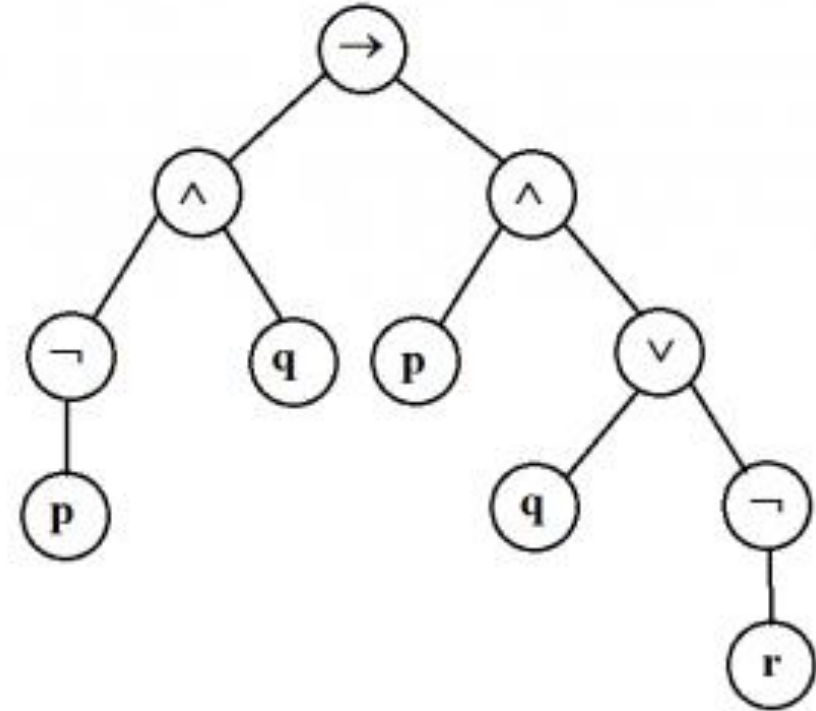
- Parse trees are a way to write down formulas
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- The higher a node, the more priority it has!



Parse Trees

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How to reconstruct a formula from a parse tree?

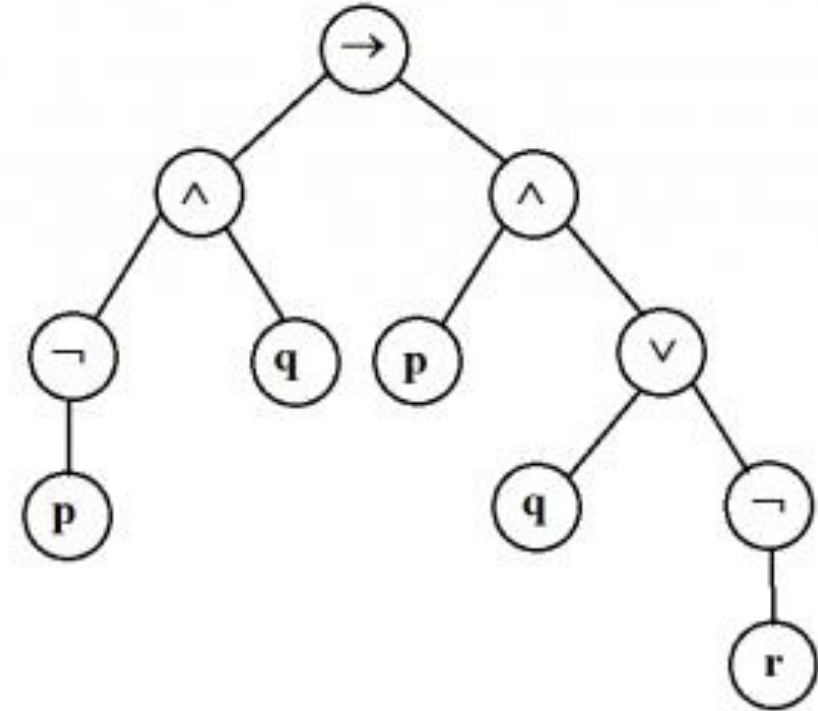


Parse Trees

- Parse trees are a way to write down formulas
- They look like this:
- The higher a node, the more priority it has!

How to reconstruct a formula from a parse tree?

- 1) Start with the connective with highest priority
- 2) Write down the nodes that belong to it on either side
 - Is a node a connective? Repeat step 1



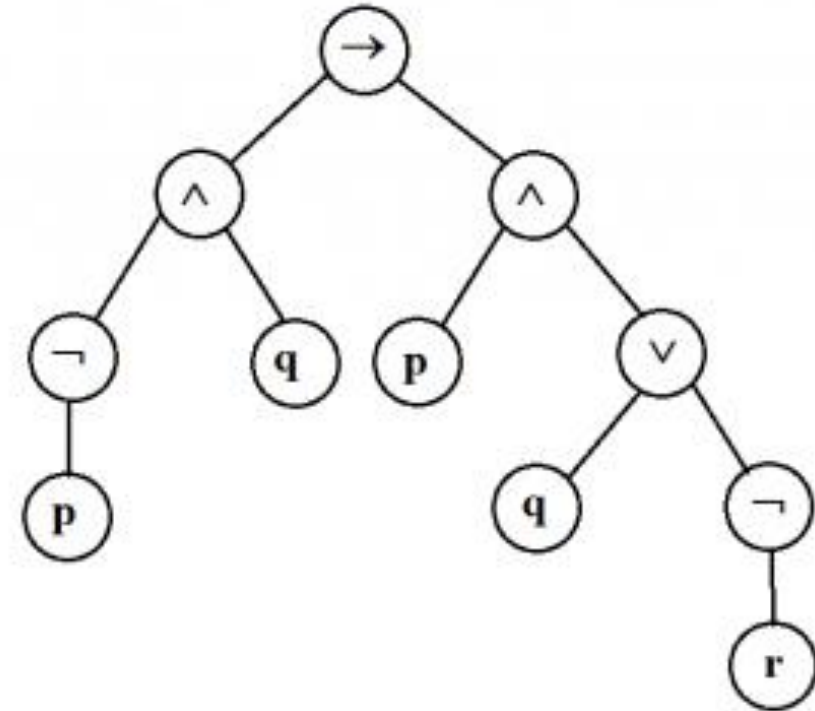
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Let's try it on the tree to the right!



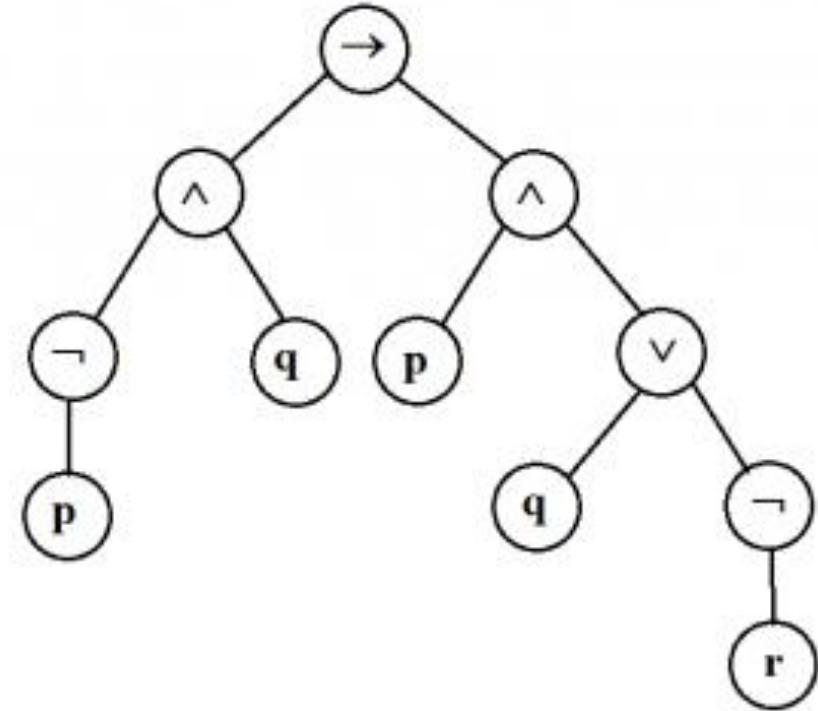
$$(\neg p \wedge q) \rightarrow (p \wedge (q \vee \neg r))$$

Parse Trees

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- The higher a node, the more priority it has!

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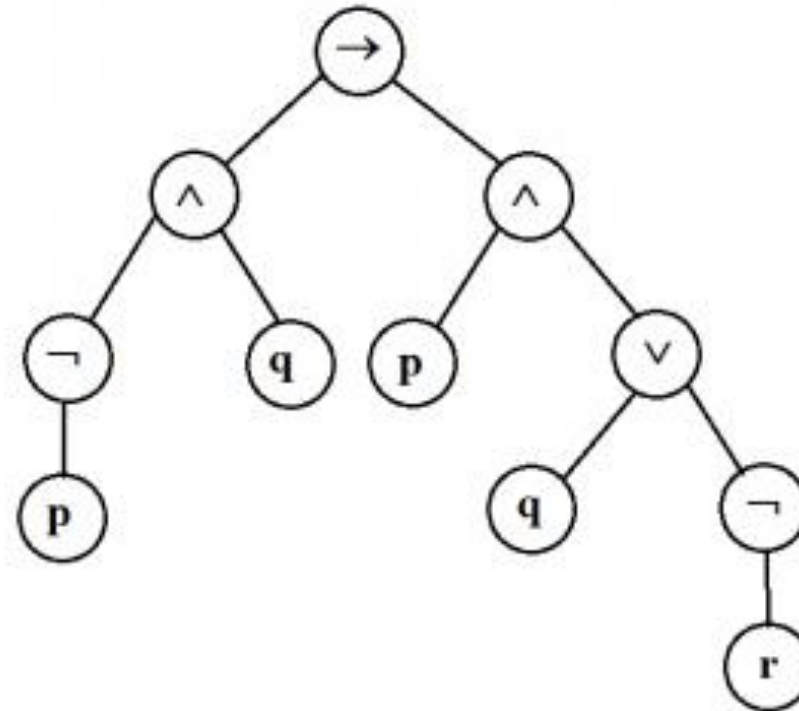
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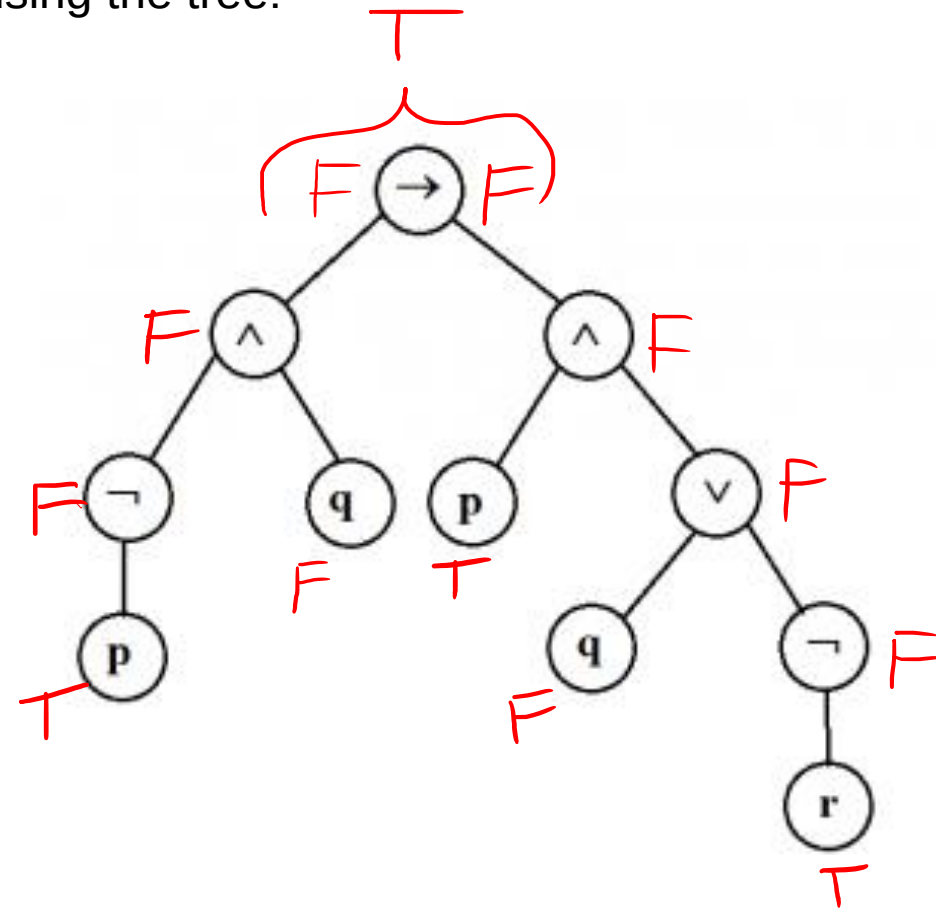
Parse Trees

- Truth value of the formula can be computed using the tree!
- Bottom-up evaluation
- Write down the truth value of each node, starting at the bottom



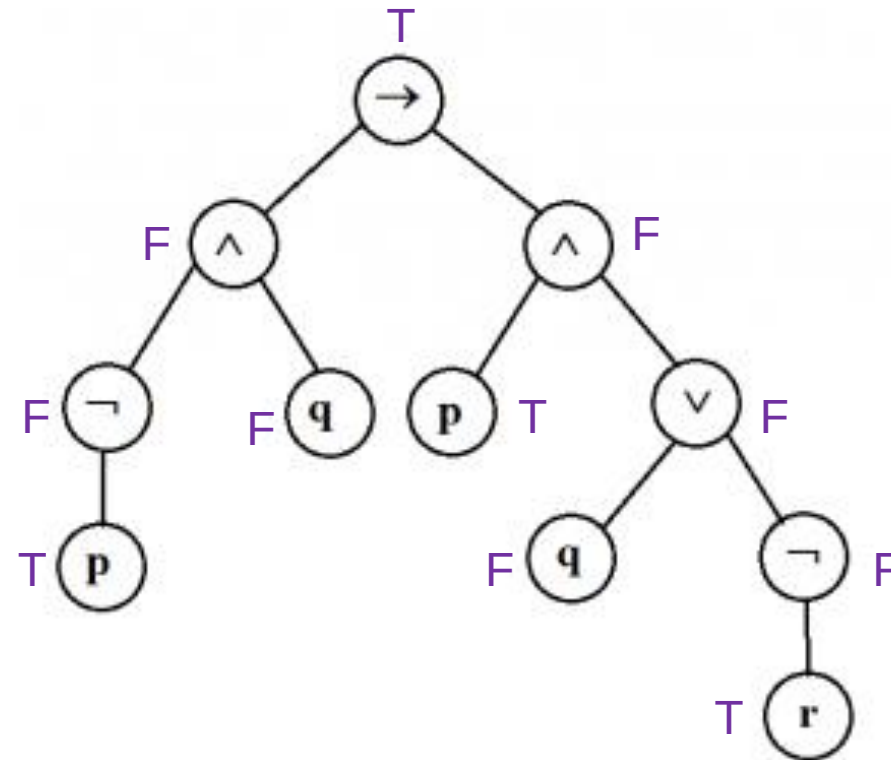
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- Truth value of the formula can be computed using the tree!
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- Write down the truth value of each node, starting at the bottom
- Let's try it!
- Start with $p = T$, $q = F$, $r = T$



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- Truth value of the formula can be computed using the tree!
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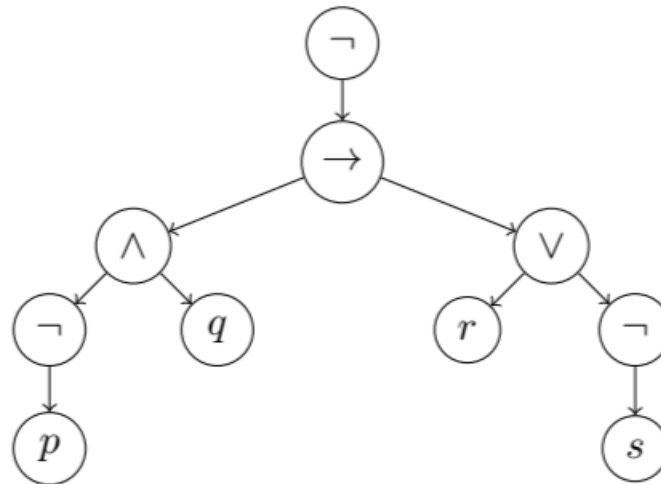


Parse Trees: Exercise!

Exam question: (resit 2019)

1. Parse tree (5+6 points)

- (a) For the following parse tree, find the propositional logic formula that it represents.



How to reconstruct a formula from a parse tree?

1) Start with the connective with highest priority

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- Is a node a connective? Repeat step 1

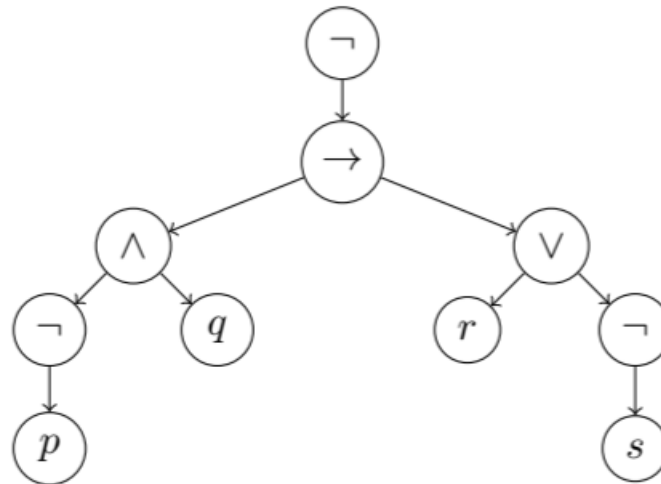
$$\neg ((\neg p \wedge q) \rightarrow (r \vee \neg s))$$

Parse Trees: Exercise!

Exam question: (resit 2019)

1. Parse tree (5+6 points)

- (a) For the following parse tree, find the propositional logic formula that it represents.



Answer: $\neg(\neg p \wedge q) \rightarrow (r \vee \neg s)$

How to reconstruct a formula from a parse tree?

1) Start with the connective with highest priority

2) Write down the nodes that belong to it on either side

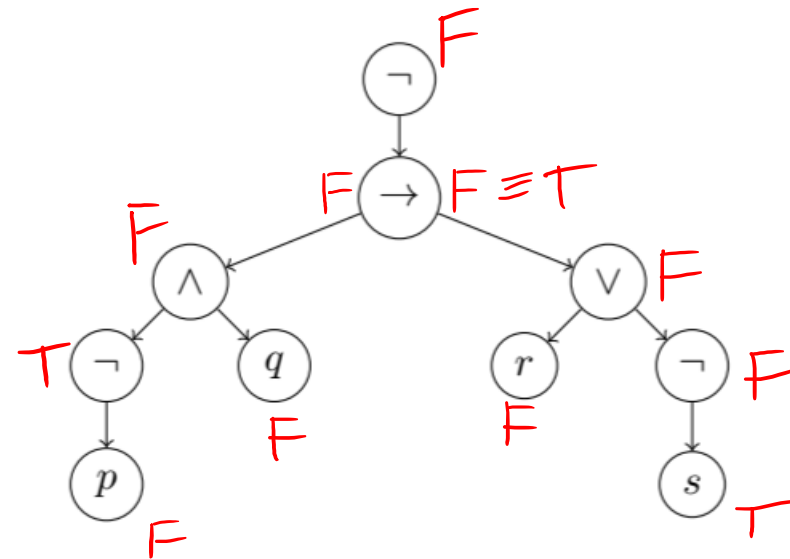
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Parse Trees: Exercise!

Exam question: (resit 2019)

1. Parse tree (5+6 points)

~~(a) For the following parse tree, find the propositional logic formula that it represents.~~



(b) Draw how the truth value of this formula is determined by means of its parse tree in a bottom-up fashion, if $p = F$, $q = F$, $r = F$, and $s = T$.

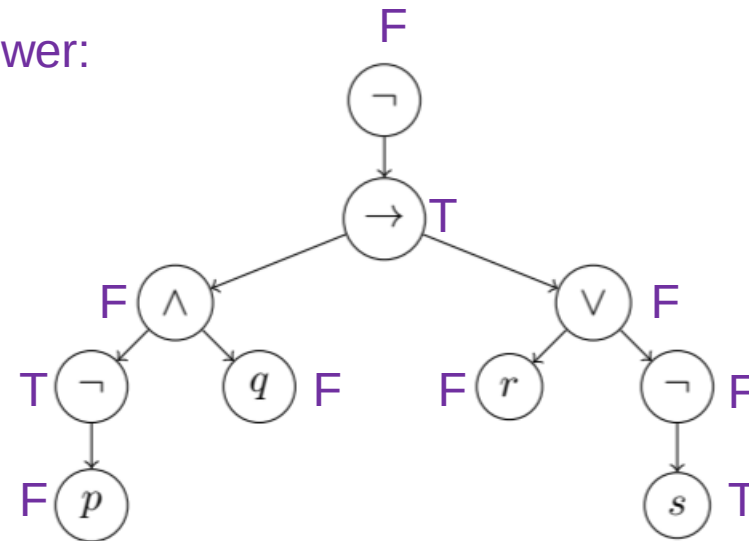
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Answer:



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Parse Trees

Other way around!

How to construct a parse tree from a formula?

Parse Trees

Other way around!

How to construct a parse tree from a formula?

1. Find the connective with the highest priority in the formula and put this at the top of the tree
2. Look at the variables left and right of the connective. If they contain another variable, restart step 1 for this variable

Parse Trees: Example

Other way around!

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Formula: $((p \wedge q) \rightarrow (r \vee \neg p)) \vee q$

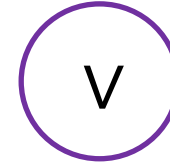
() ∨ ()

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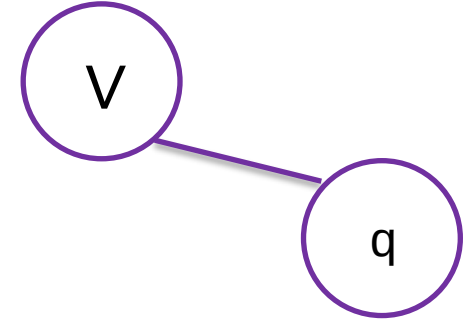
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$(\quad) \rightarrow (\quad)$



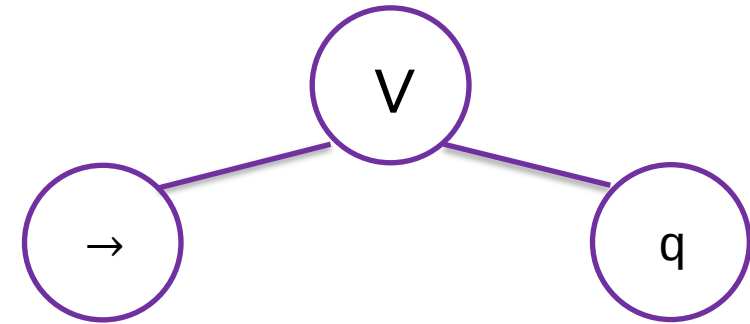
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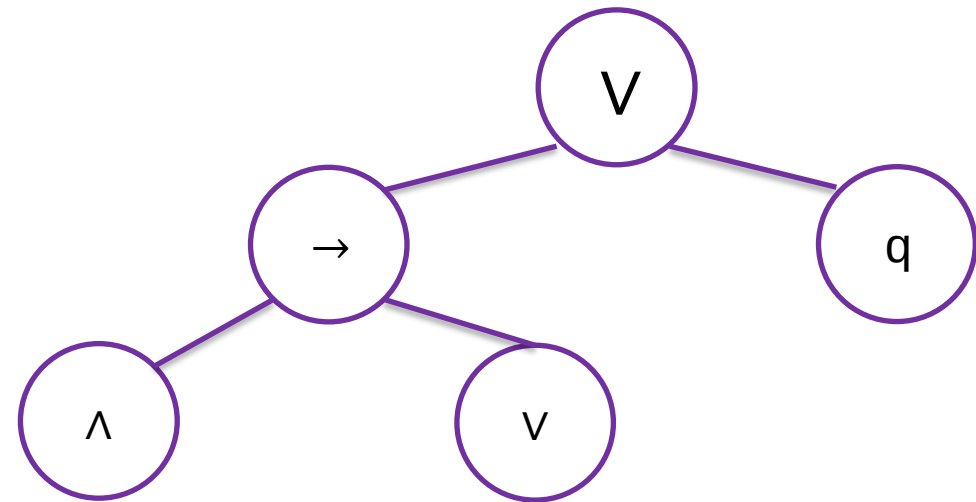
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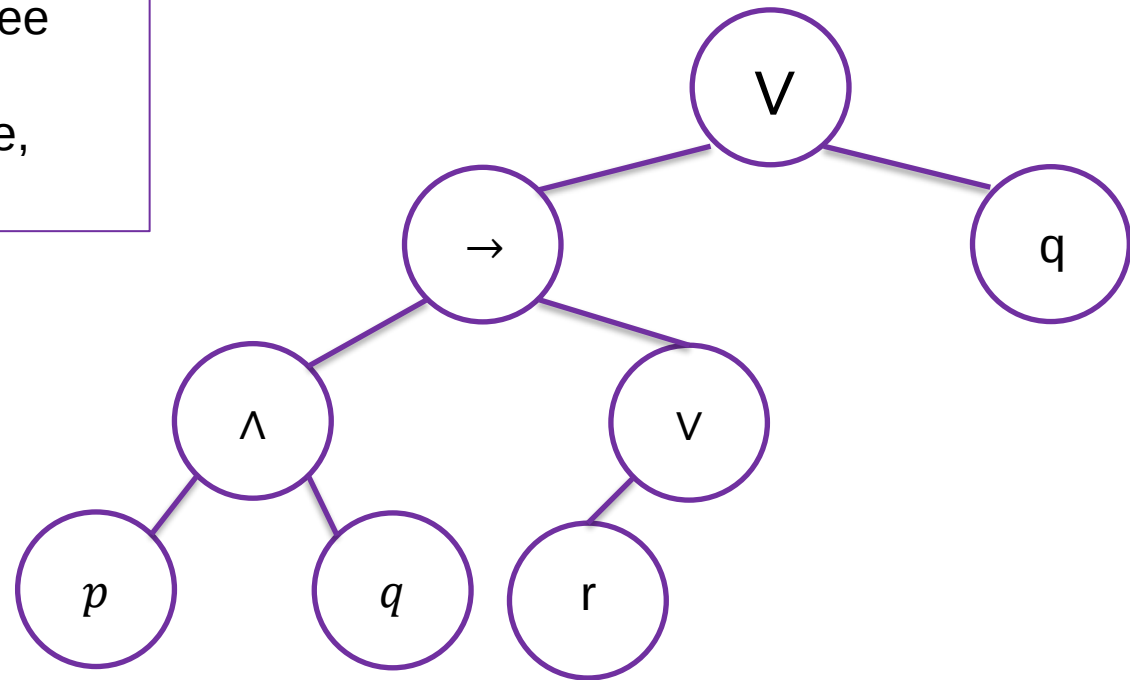
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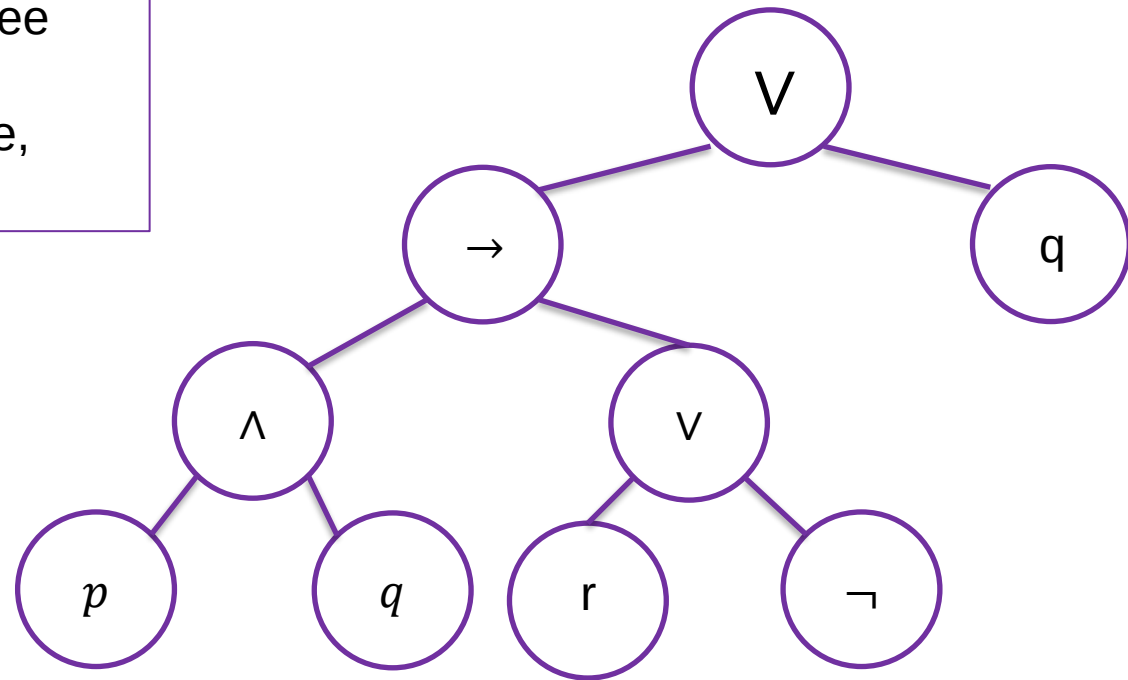
Parse Trees: Example

Other way around!

How to construct a parse tree from a formula?

1. Find the connective with the highest priority in the formula and put this at the top of the tree
2. Look at the variables left and right of the connective. If they contain another variable, restart step 1 for this variable

Formula: $((p \wedge q) \rightarrow (r \vee \neg p)) \vee q$



Parse Trees: Example

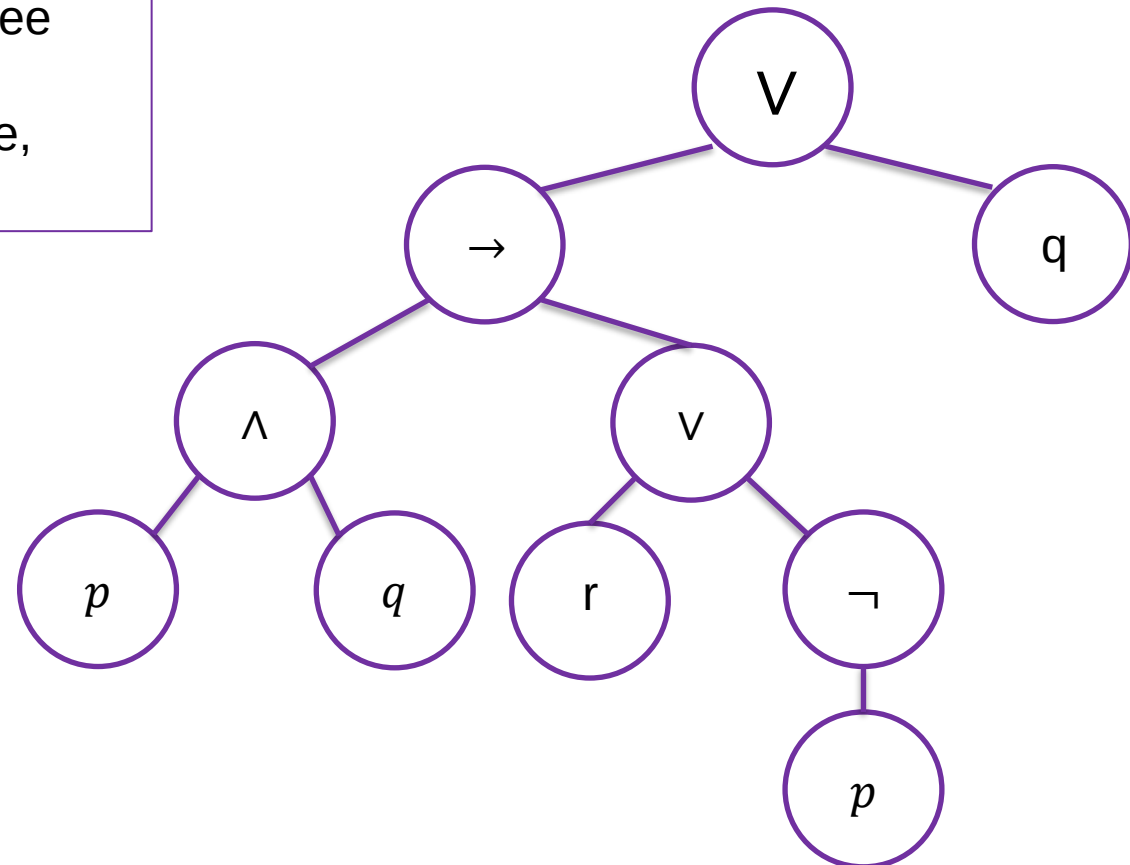
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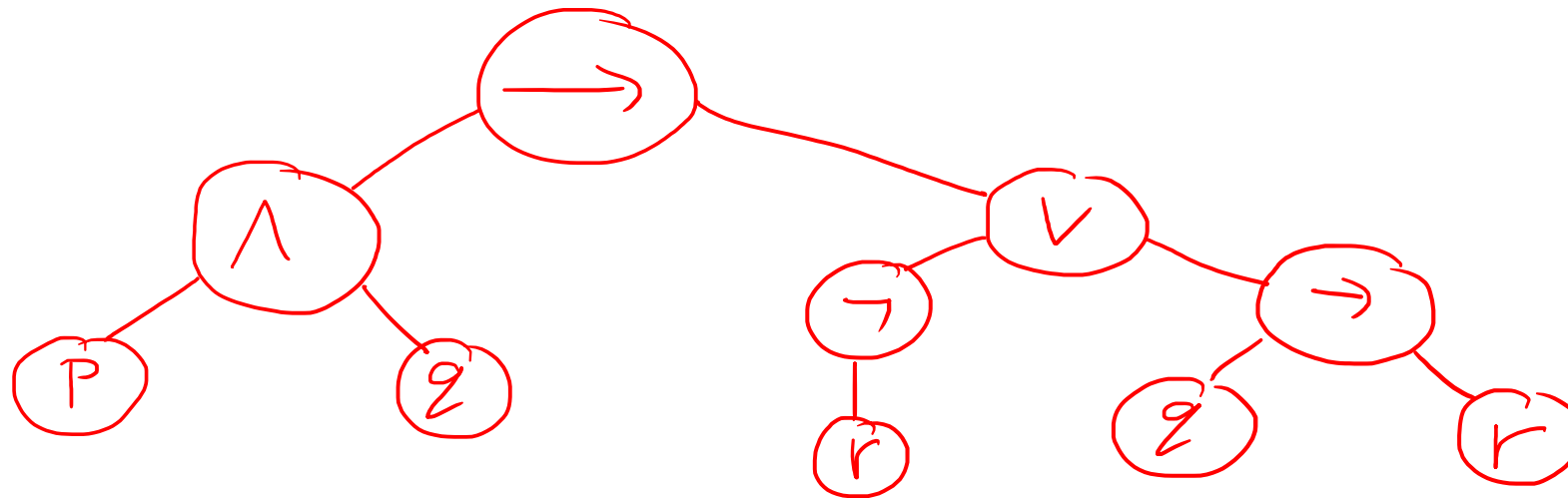
$(() \wedge ()) \rightarrow (() \vee ())$



Parse Trees: Exercise!

Draw the parse tree of the following formula:

$$(p \wedge q) \rightarrow (\neg r \vee (q \rightarrow r))$$



How to construct a parse tree from a formula?

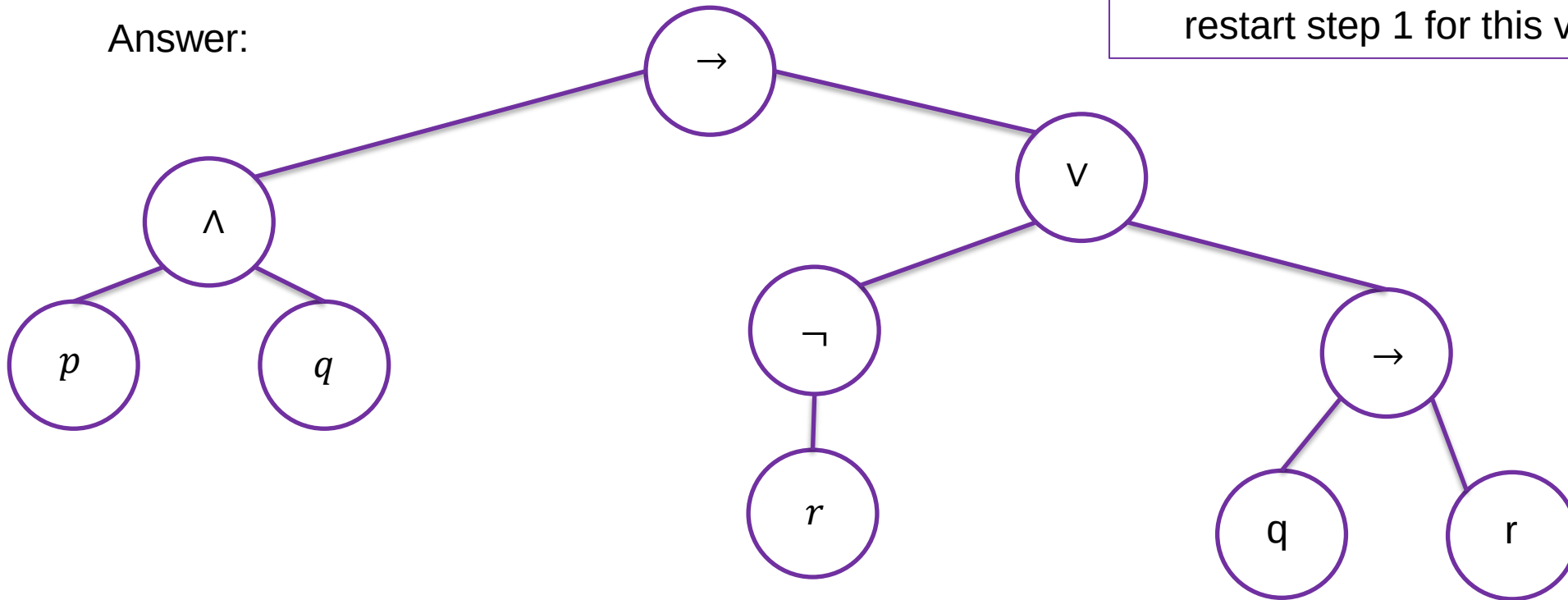
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Parse Trees: Exercise!

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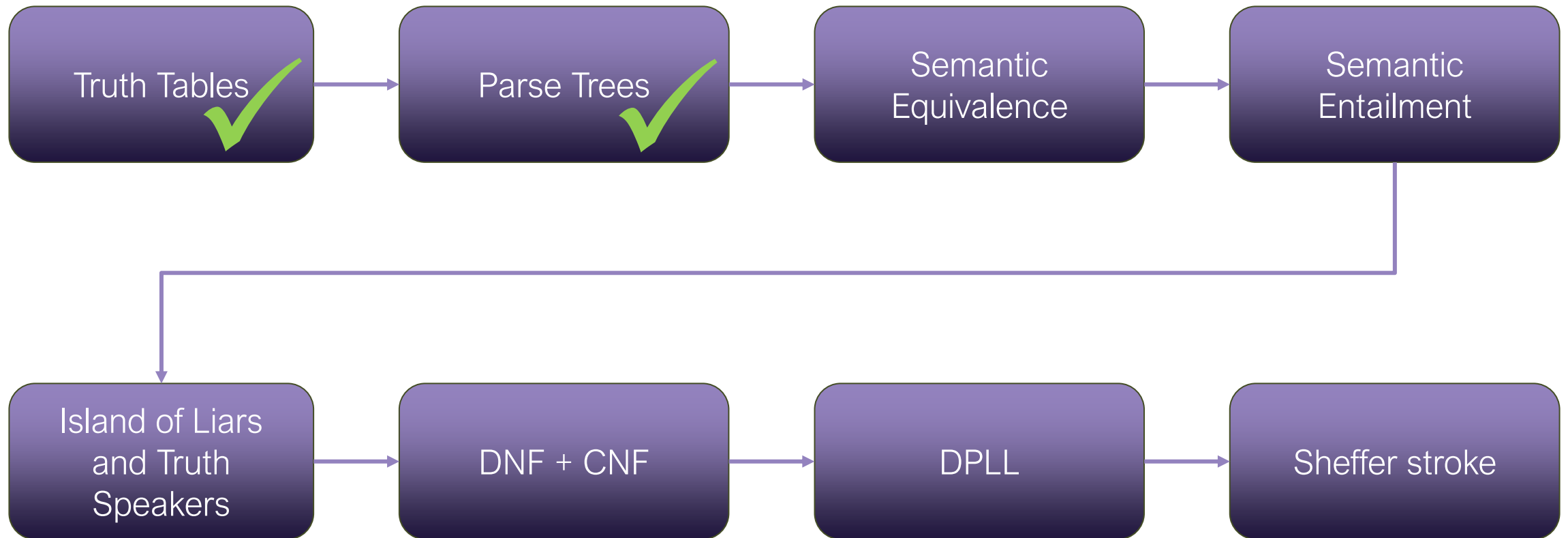
Answer:



How to construct a parse tree from a formula?

1. Find the connective with the highest priority in the formula and put this at the top of the tree
2. Look at the variables left and right of the connective. If they contain another variable, restart step 1 for this variable

Planning Day 1



Semantic Equivalence

Semantic Equivalence

Definition:

Formulas ϕ and ψ are **semantically equivalent**,
if each valuation that makes $\phi_1, \dots, \phi_n$ true, also makes ψ true.

Notation: $\phi \equiv \psi$

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

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if each valuation that makes $\phi_1, \dots, \phi_n$ true, also makes ψ true.

Notation: $\phi \equiv \psi$

$$p \rightarrow q \equiv \neg p \vee q$$

p	q	$p \rightarrow q$	$\neg p$	$\neg p \vee q$
T	T	T	F	T
T	F	F	F	F
F	T	T	T	T
F	F	T	T	T

If $\phi \equiv \psi$, then ϕ and ψ have the same truth table.

So ϕ and ψ are T on exactly the same rows.

the same!

Semantic Equivalence

With semantic equivalence, we can express certain connectives in another way:

Commutativity:

$$\phi \vee \psi \equiv \psi \vee \phi$$

$$\phi \wedge \psi \equiv \psi \wedge \phi$$

Associativity:

$$\phi \vee (\psi \vee \chi) \equiv (\phi \vee \psi) \vee \chi$$

$$\phi \wedge (\psi \wedge \chi) \equiv (\phi \wedge \psi) \wedge \chi$$

Distributivity:

$$\phi \vee (\psi \wedge \chi) \equiv (\phi \vee \psi) \wedge (\phi \vee \chi)$$

$$\phi \wedge (\psi \vee \chi) \equiv (\phi \wedge \psi) \vee (\phi \wedge \chi)$$

Identities:

$$\phi \vee \perp \equiv \phi$$

$$\phi \wedge \top \equiv \phi$$

Domination:

$$\phi \vee \top \equiv \top$$

$$\phi \wedge \perp \equiv \perp$$

Idempotence:

$$\phi \vee \phi \equiv \phi$$

$$\phi \wedge \phi \equiv \phi$$

Complement:

$$\phi \vee \neg\phi \equiv \top$$

$$\phi \wedge \neg\phi \equiv \perp$$

De Morgan:

$$\neg(\phi \vee \psi) \equiv \neg\phi \wedge \neg\psi$$

$$\neg(\phi \wedge \psi) \equiv \neg\phi \vee \neg\psi$$

Involution:

$$\neg\neg\phi \equiv \phi$$

Semantic Equivalence: Exercise!

Exam Question: (Resit 2020)

1. Distributivity laws (6 + 6 points)

For each of the two distributivity laws below, regarding conjunction and exclusive or, either argue that it holds or give a counterexample to show that it does not hold.

(a) $(x \wedge y) \oplus z \equiv (x \oplus z) \wedge (y \oplus z)$

Semantic Equivalence: Exercise!

Exam Question: (Resit 2020)

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(a) $(x \wedge y) \oplus z \equiv (x \oplus z) \wedge (y \oplus z)$

x	y	z	$x \wedge y$	$(x \wedge y) \oplus z$	$x \oplus z$	$y \oplus z$	$\phi \wedge \psi$
T	T	T	T	F	F	F	F
T	T	F	T	T	T	T	T
T	F	T	F	T	F	T	F
T	F	F	F	F	T	F	F
F	T	T	F	T	T	F	F
F	T	F	F	F	F	T	F
F	F	T	F	T	T	T	T
F	F	F	F	F	F	F	F

$\oplus$
 $\left. \begin{matrix} T, F \\ F, T \end{matrix} \right\} T$
 Counterexample:

$x = T$
 $y = F$
 $z = T$

Semantic Equivalence: Exercise!

Exam Question: (Resit 2020)

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For each of the two distributivity laws below, regarding conjunction and exclusive or, either argue that it holds or give a counterexample to show that it does not hold.

$$(a) (x \wedge y) \oplus z \equiv (x \oplus z) \wedge (y \oplus z)$$

Answer:

x	y	z	$x \wedge y$	$(x \wedge y) \oplus z$	$x \oplus z$	$y \oplus z$	$(x \oplus z) \wedge (y \oplus z)$
T	T	T	T	F	F	F	F
T	T	F	T	T	T	T	T
T	F	T	F	T	F	T	F
T	F	F	F	F	T	F	F
F	T	T	F	T	T	F	F
F	T	F	F	F	F	T	F
F	F	T	F	T	T	T	T
F	F	F	F	F	F	F	F

The two columns of the truth table do not match, so it does not hold.

Semantic Equivalence: Exercise!

Exam Question: (Resit 2020)

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For each of the two distributivity laws below, regarding conjunction and exclusive or, either argue that it holds or give a counterexample to show that it does not hold.

$$(a) (x \wedge y) \oplus z \equiv (x \oplus z) \wedge (y \oplus z)$$

F1

F2

F1

F2

Answer:

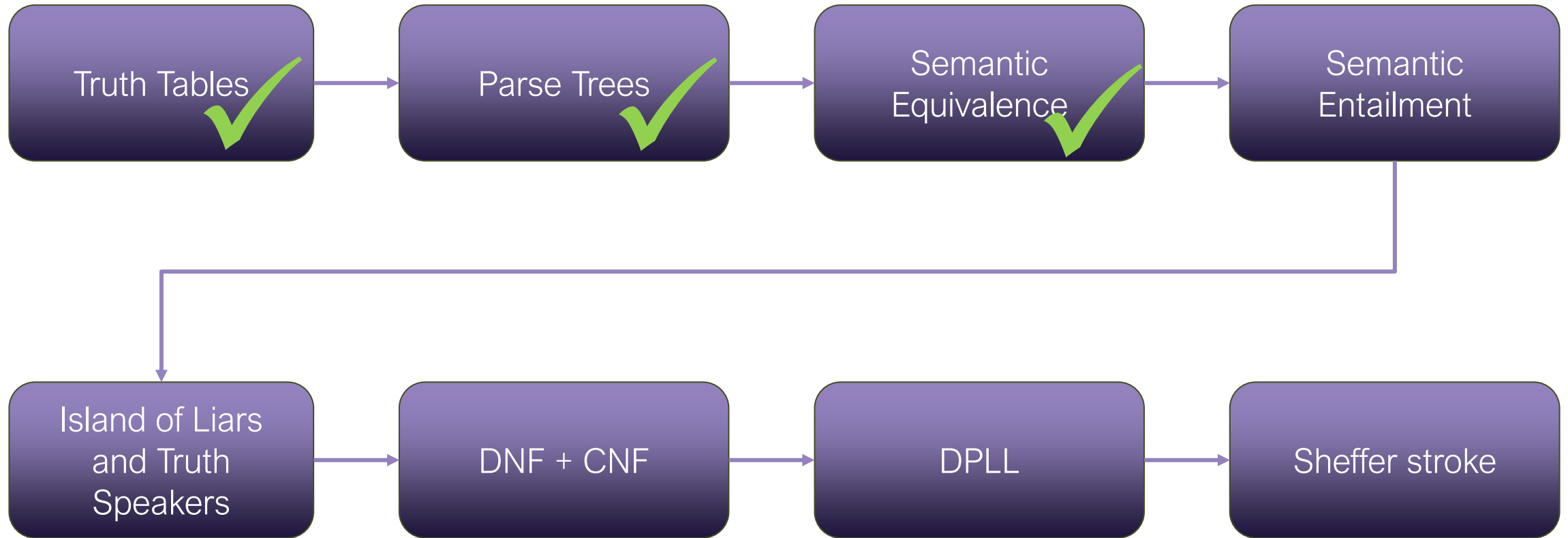
x	y	z	$x \wedge y$	$(x \wedge y) \oplus z$	$x \oplus z$	$y \oplus z$	$(x \oplus z) \wedge (y \oplus z)$
T	T	T	T	F	F	F	F
T	T	F	T	T	T	T	T
T	F	T	F	T	F	T	F
T	F	F	F	F	T	F	F
F	T	T	F	T	T	F	F
F	T	F	F	F	F	T	F
F	F	T	F	T	T	T	T
F	F	F	F	F	F	F	F

The two columns of the truth table do not match, so it does not hold.

A **counterexample** would be any valuation where the values do not match.
e.g. if $x = T, y = F, z = T$

→ Row 3 where
 $F1 = T, F2 = F$

Planning Day 1



Break!

Semantic Entailment

Semantic Entailment

Definition:

A formula ψ is **semantically entailed** by formulas $\phi_1, \dots, \phi_n$ if each valuation that makes $\phi_1, \dots, \phi_n$ true, also makes ψ true

Notation: $\phi_1, \dots, \phi_n \models \psi$

(Each valuation corresponds to a row in the truth table)

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Notation: $\phi_1, \dots, \phi_n \models \psi$

(Each valuation corresponds to a row in the truth table)

Does $p \rightarrow q, \neg q \models \neg p$ hold?

p	q	$p \rightarrow q$	$\neg q$	$\neg p$
T	T	T	F	F
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T



So in each row where both *premises* are true, the *conclusion* is also true.

Therefore, the semantic entailment is **valid**.

Does $p \vee q \models p \rightarrow q$ hold?

p	q	$p \vee q$	$p \rightarrow q$
T	T	T	T
T	F	T	F
F	T	T	T
F	F	F	T



This shows that the semantic entailment is **not valid**:

The valuation $p \mapsto \text{T}, q \mapsto \text{F}$ (row 2) is a **counterexample**.

Semantic Entailment

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Notation: $\phi_1, \dots, \phi_n \models \psi$

(Each valuation corresponds to a row in the truth table)

$\models \phi$ denotes that ϕ is a *tautology*.

If $\phi_1 \models \psi$, then also $\phi_1, \phi_2 \models \psi$.

Semantic Entailment

Definition:

A formula ψ is **semantically entailed** by formulas $\phi_1, \dots, \phi_n$ if each valuation that makes $\phi_1, \dots, \phi_n$ true, also makes ψ true

Notation: $\phi_1, \dots, \phi_n \models \psi$

(Each valuation corresponds to a row in the truth table)

Let's check if

$p \wedge \neg q \rightarrow r, \neg r, p \models q$
is valid

p	q	r	$\neg q$	$\neg r$	$p \wedge \neg q$	$p \wedge \neg q \rightarrow r$
T	T	T	F	F	F	T
T	T	F	F	T	F	T
T	F	T	T	F	T	T
T	F	F	T	T	T	F
F	T	T	F	F	F	T
F	T	F	F	T	F	T
F	F	T	T	F	F	T
F	F	F	T	T	F	T

Semantic Entailment

Definition:

A formula ψ is **semantically entailed** by formulas $\phi_1, \dots, \phi_n$ if each valuation that makes $\phi_1, \dots, \phi_n$ true, also makes ψ true

Notation: $\phi_1, \dots, \phi_n \models \psi$

(Each valuation corresponds to a row in the truth table)

Let's check if

$p \wedge \neg q \rightarrow r, \neg r, p \models q$ ✓
is valid

As you can see,
wherever the premises
are true, the conclusion
is also true.

So we can say that q is
semantically entailed by

$p \wedge \neg q \rightarrow r, \neg r, p$

p	q	r	$\neg q$	$\neg r$	$p \wedge \neg q$	$p \wedge \neg q \rightarrow r$
T	T	T	F	F	F	T
T	T	F	F	T	F	T
T	F	T	T	F	T	T
T	F	F	T	T	T	F
F	T	T	F	F	F	T
F	T	F	F	T	F	T
F	F	T	T	F	F	T
F	F	F	T	T	F	T

Semantic Entailment

Definition:

A formula ψ is **semantically entailed** by formulas $\phi_1, \dots, \phi_n$ if each valuation that makes $\phi_1, \dots, \phi_n$ true, also makes ψ true

Notation: $\phi_1, \dots, \phi_n \models \psi$

(Each valuation corresponds to a row in the truth table)

You can show that $\phi_1, \dots, \phi_n \models \psi$ does not hold, by giving a **counterexample!**



Find a valuation for which $\phi_1, \dots, \phi_n$ is true, but ψ is not true

Semantic Entailment

Definition:

A formula ψ is **semantically entailed** by formulas $\phi_1, \dots, \phi_n$ if each valuation that makes $\phi_1, \dots, \phi_n$ true, also makes ψ true

Notation: $\phi_1, \dots, \phi_n \models \psi$

(Each valuation corresponds to a row in the truth table)

$\phi \equiv \psi$ if and only if (iff) $\phi \models \psi$ and $\psi \models \phi$

↳ truth tables
will be the
same

Semantic Entailment: Exercise!

Exam Question: (Midterm 2018)

1. Semantic entailment (5 + 5 points)

Argue for each of the following two semantic entailments whether it holds. If so, show this by means of a truth table; if not, give a concrete counterexample.

(a) $(\phi \oplus \psi) \oplus \chi \models \phi \oplus (\psi \oplus \chi)$

~~(b) $\phi \rightarrow (\psi \rightarrow \chi) \models (\phi \rightarrow \psi) \rightarrow \chi$~~

Definition:

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Semantic Entailment: Exercise!

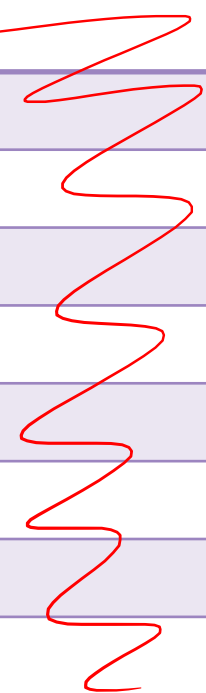
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(a) $(\phi \oplus \psi) \oplus \chi \models \phi \oplus (\psi \oplus \chi)$ F1 F2

~~(b) $\phi \rightarrow (\psi \rightarrow \chi) \models (\phi \rightarrow \psi) \rightarrow \chi$~~

ϕ	ψ	χ	$\phi \oplus \psi$	$(\phi \oplus \psi) \oplus \chi$	$\psi \oplus \chi$	$\phi \oplus \alpha$	
T	T	T	F	T ✓	F	T ✓	
T	T	F	F	F	T	F	
T	F	T	T	F	T	F	
T	F	F	T	T ✓	F	T ✓	
F	T	T	T	F	F	F	
F	T	F	T	T ✓	T	T ✓	
F	F	T	F	T ✓	T	T ✓	
F	F	F	F	F	F	F	

Definition:

A formula ψ is **semantically entailed** by formulas $\phi_1, \dots, \phi_n$ if each valuation that makes $\phi_1, \dots, \phi_n$ true, also makes ψ true

Notation: $\phi_1, \dots, \phi_n \models \psi$

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~~(b) $\phi \rightarrow (\psi \rightarrow \chi) \models (\phi \rightarrow \psi) \rightarrow \chi$~~

$F1 \models F2$
 $F2 \models F1$ } $F1 \equiv F2$

Answer:

ϕ	ψ	χ	$\phi \oplus \psi$	$(\phi \oplus \psi) \oplus \chi$	$\psi \oplus \chi$	$\phi \oplus (\psi \oplus \chi)$
T	T	T	F	T	F	T ✓
T	T	F	F	F	T	F
T	F	T	T	F	T	F
T	F	F	T	T	F	T ✓
F	T	T	T	F	F	F
F	T	F	T	T	T	T ✓
F	F	T	F	T	T	T ✓
F	F	F	F	F	F	F

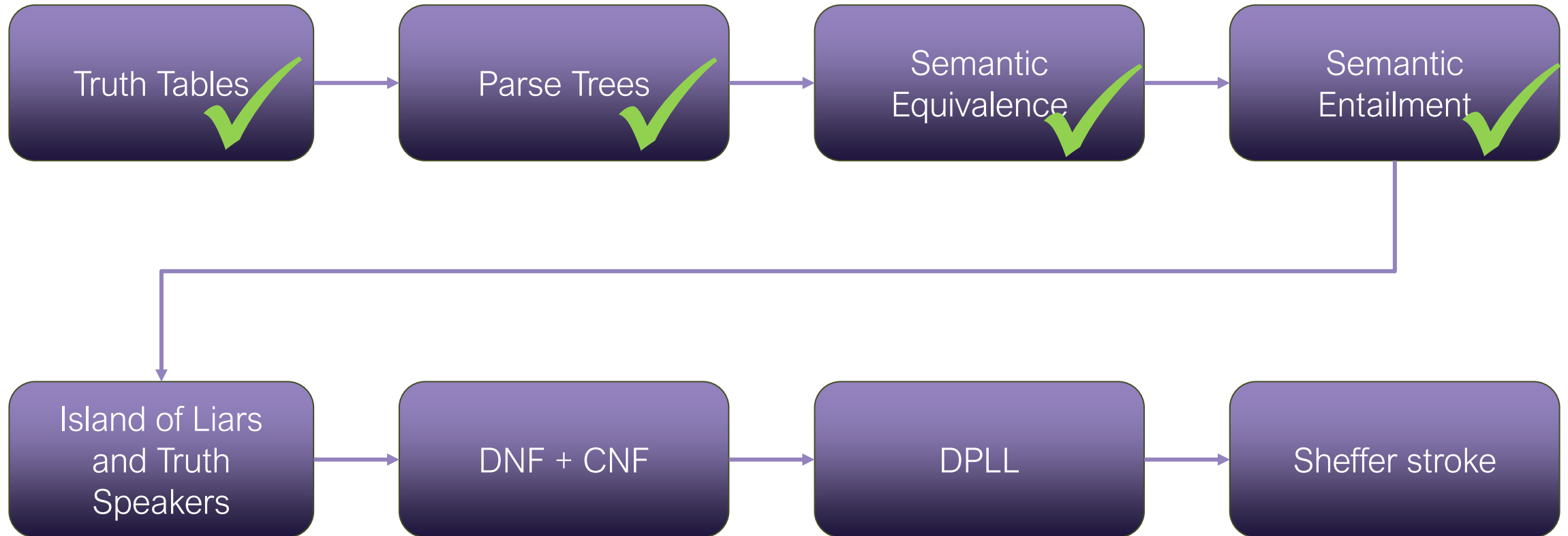
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Notation: $\phi_1, \dots, \phi_n \models \psi$

The semantic entailment holds, because in every row where the fifth column contains $\mathbb{T}$, the seventh column also contains $\mathbb{T}$.

Planning Day 1



Island of Liars and Truth Speakers

Island of Liars and Truth Speakers

On the *Island of Liars and Truth Speakers*, every inhabitant has the peculiar property of either:

- ▶ always lying, or
- ▶ always speaking the truth

They only utter declarative sentences, and answer 'yes' or 'no'.

For each islander x , the propositional variable t_x expresses:

x is a truth speaker



$\neg t_x \rightarrow x$ is a liar

Island of Liars and Truth Speakers

If an islander x makes an assertion ϕ , then:

- ▶ ϕ is true if x is a truth speaker
- ▶ ϕ is false if x is a liar

Hence, irrespective of whether or not x is a truth speaker, the following bi-implication is true:

$$t_x \leftrightarrow \phi \Rightarrow T \text{ iff the islander is a truth speaker}$$
$$t_x \leftrightarrow \phi$$
$$T \leftrightarrow T \Rightarrow T$$

Island of Liars and Truth Speakers

If an islander x makes an assertion ϕ , then:

- ▶ ϕ is true if x is a truth speaker
- ▶ ϕ is false if x is a liar

Hence, irrespective of whether or not x is a truth speaker, the following bi-implication is true:

$$t_x \leftrightarrow \phi$$

Puzzle

You meet two islanders a and b .

a says: "We are both liars."

What are a and b ?

a says, in propositional logic: $\neg t_a \wedge \neg t_b$.

So $t_a \leftrightarrow \neg t_a \wedge \neg t_b$ is true.

$t_a \leftrightarrow \neg t_a \wedge \neg t_b$: assume a is a truth speaker

$T \leftrightarrow \neg T \wedge \neg T$

$T \leftrightarrow F \wedge F$

$T \leftrightarrow F \longrightarrow F$, so a is a liar, b truth

Island of Liars and Truth Speakers

Puzzle

You meet two islanders *a* and *b*.

a says: “*We are both liars.*”

What are *a* and *b*?

a says, in propositional logic: $\neg t_a \wedge \neg t_b$.

So $t_a \leftrightarrow \neg t_a \wedge \neg t_b$ is true.

t_a	t_b	$\neg t_a$	$\neg t_b$	$\neg t_a \wedge \neg t_b$	$t_a \leftrightarrow \neg t_a \wedge \neg t_b$
T	T	F	F	F	F
T	F	F	T	F	F
F	T	T	F	F	T
F	F	T	T	T	F

Steps to solve this:

1. Make a truth table containing
 - the truth value of all islanders ($t_a, t_b, t_c, \dots$) and
 - the bi-implication of all statements they make ($t_a \leftrightarrow \text{statement } a$)
2. Check in which rows all bi-implications are true
3. Use this row to see if the islanders are truth speakers or liars

Island of Liars and Truth Speakers

Puzzle

You meet two islanders *a* and *b*.

a says: "We are both liars."

What are *a* and *b*?

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So $t_a \leftrightarrow \neg t_a \wedge \neg t_b$ is true.

t_a	t_b	$\neg t_a$	$\neg t_b$	$\neg t_a \wedge \neg t_b$	$t_a \leftrightarrow \neg t_a \wedge \neg t_b$
T	T	F	F	F	F
T	F	F	T	F	F
F	T	T	F	F	T
F	F	T	T	T	F

What is the conclusion here?

Steps to solve this:

1. Make a truth table containing
 - the truth value of all islanders ($t_a, t_b, t_c, \dots$) and
 - the bi-implication of all statements they make ($t_a \leftrightarrow \text{statement } a$)
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Island of Liars and Truth Speakers

Puzzle

You meet two islanders *a* and *b*.

a says: "We are both liars."

What are *a* and *b*?

a says, in propositional logic: $\neg t_a \wedge \neg t_b$.

So $t_a \leftrightarrow \neg t_a \wedge \neg t_b$ is true.

t_a	t_b	$\neg t_a$	$\neg t_b$	$\neg t_a \wedge \neg t_b$	$t_a \leftrightarrow \neg t_a \wedge \neg t_b$
T	T	F	F	F	F
T	F	F	T	F	F
F	T	T	F	F	T
F	F	T	T	T	F

A is a liar and B is a truthspeaker!

Steps to solve this:

1. Make a truth table containing
 - the truth value of all islanders ($t_a, t_b, t_c, \dots$) and
 - the bi-implication of all statements they make ($t_a \leftrightarrow \text{statement } a$)
2. Check in which rows all bi-implications are true
3. Use this row to see if the islanders are truth speakers or liars

Island of Liars and Truth Speakers: Exercise!

Exam question: (midterm 2021)

Island puzzle (12 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet islanders A and B .

A says: "If I am a liar, then B is a truth speaker."

B says: "Exactly one of us two is a liar."

Determine via a truth table for both A and B whether this islander is a truth speaker or a liar.

t_a	t_b	$\neg t_a \rightarrow t_b$	$t_a \oplus t_b$	$t_a \leftrightarrow (\neg t_a \rightarrow t_b)$	$t_b \leftrightarrow (t_a \oplus t_b)$
T	T				
T	F				
F	T				
F	F				

Steps to solve this:

1. Make a truth table containing
 - the truth value of all islanders ($t_a, t_b, t_c, \dots$) and
 - the bi-implication of all statements they make ($t_a \leftrightarrow \text{statement } a$)
2. Check in which rows all bi-implications are true
3. Use this row to see if the islanders are truth speakers or liars

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t_a	t_b	$\neg t_a \rightarrow t_b$	$t_a \oplus t_b$	$t_a \leftrightarrow (\neg t_a \rightarrow t_b)$	$t_b \leftrightarrow (t_a \oplus t_b)$
T	T	T	F	T	F
T	F	T	T	T	F
F	T	T	T	F	T
F	F	F	F	T	T

→ A and B are both liars

Steps to solve this:

1. Make a truth table containing
 - the truth value of all islanders ($t_a, t_b, t_c, \dots$) and
 - the bi-implication of all statements they make ($t_a \leftrightarrow \text{statement } a$)
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Island of Liars and Truth Speakers: Exercise!

Exam question: (midterm 2021)

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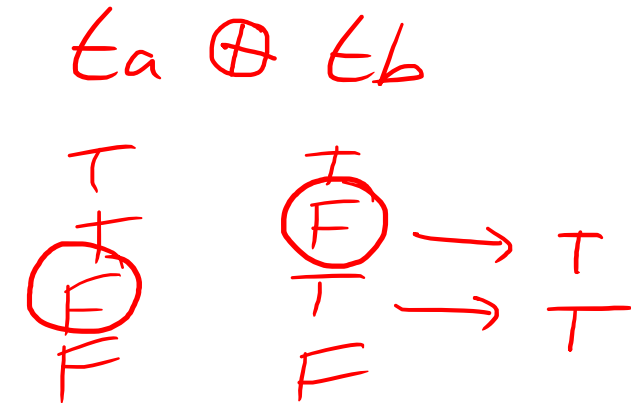
A says: "If I am a liar, then B is a truth speaker."

B says: "Exactly one of us two is a liar." $\rightarrow$ $A = T, B = F$ $A = F, B = T$

Determine via a truth table for both A and B whether this islander is a truth speaker or a liar.

Solution:

t_a	t_b	$\neg t_a \rightarrow t_b$	$t_a \oplus t_b$	$t_a \leftrightarrow (\neg t_a \rightarrow t_b)$	$t_b \leftrightarrow (t_a \oplus t_b)$
T	T	T	F	T	F
T	F	T	T	T	F
F	T	T	T	F	T
F	F	F	F	T	T



A and B are both liars

Island of Liars and Truth Speakers: Exercise!

Exam question: (final 2022)

1. Island of liars and truth speakers (12 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet three islanders A , B and C .

A says: “ B or C is a truth speaker, but not both.”

B says: “ A is a truth speaker.”

C says: “ B is a truth speaker.”

You need to determine, *by means of a truth table*, which of these three islanders speak the truth and which ones lie.

t_a	t_b	t_c	$t_b \oplus t_c$	$t_a \leftrightarrow (t_b \oplus t_c)$	$t_b \leftrightarrow t_a$	$t_c \leftrightarrow t_b$
T	T	T				
T	T	F				
T	F	T				
T	F	F				
F	T	T				
F	T	F				
F	F	T				
F	F	F				

Island of Liars and Truth Speakers: Exercise!

Exam question: (final 2022)

1. Island of liars and truth speakers (12 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet three islanders A , B and C .

A says: " B or C is a truth speaker, but not both."

B says: " A is a truth speaker."

C says: " B is a truth speaker."

You need to determine, *by means of a truth table*, which of these three islanders speak the truth and which ones lie.

t_a	t_b	t_c	$t_b \oplus t_c$	$t_a \leftrightarrow t_b \oplus t_c$	$t_b \leftrightarrow t_a$	$t_c \leftrightarrow t_b$
T	T	T	F	F	T	T
T	T	F	T	T	T	F
T	F	T	T	T	F	F
T	F	F	F	F	F	T
F	T	T	F	T	F	T
F	T	F	T	F	F	F
F	F	T	T	F	T	F
F	F	F	F	T	T	T

A, B, C are all liars

Island of Liars and Truth Speakers: Exercise!

Exam question: (final 2022)

1. Island of liars and truth speakers (12 points)

On the island of liars and truth speakers, everybody is either a liar (who always lies) or a truth speaker (who always speaks the truth). You meet three islanders A , B and C .

A says: " B or C is a truth speaker, but not both."

B says: " A is a truth speaker."

C says: " B is a truth speaker."

You need to determine, *by means of a truth table*, which of these three islanders speak the truth and which ones lie.

$$B = T$$

$$C = F$$

$$C = F$$

$$B = T$$

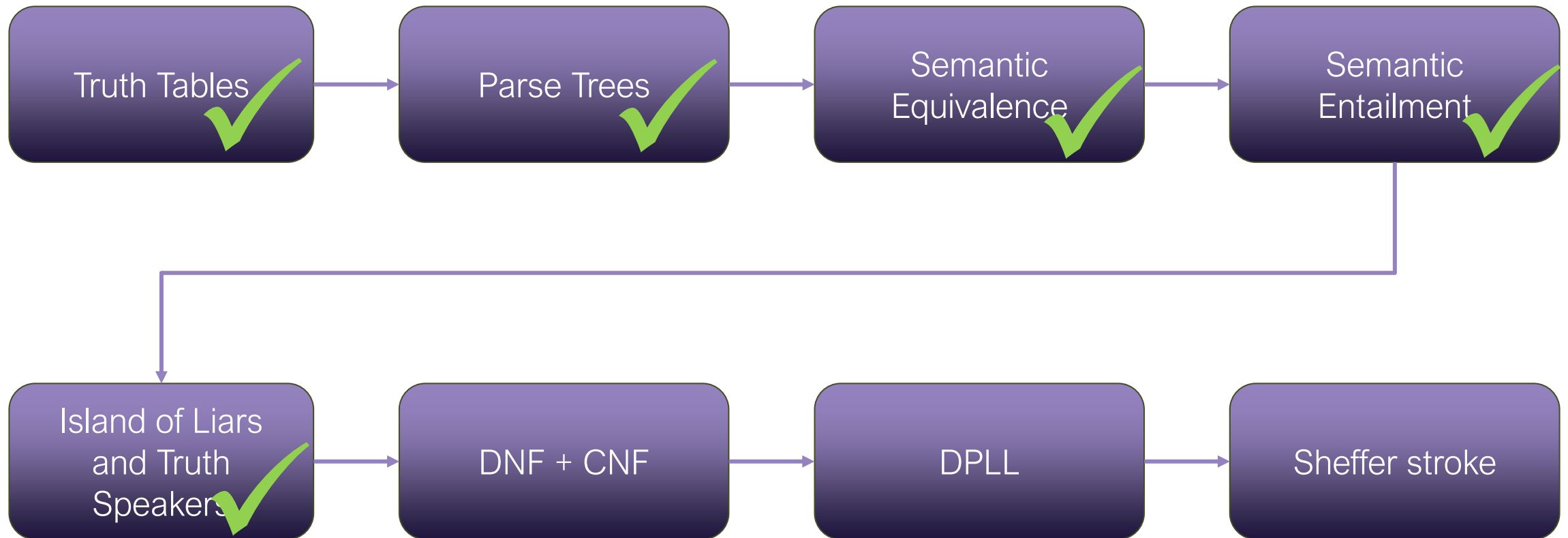
Solution:

t_a	t_b	t_c	$t_b \oplus t_c$	$t_a \leftrightarrow t_b \oplus t_c$	$t_b \leftrightarrow t_a$	$t_c \leftrightarrow t_b$
T	T	T	F	F	T	T
T	T	F	T	T	T	F
T	F	T	T	T	F	F
T	F	F	F	F	F	T
F	T	T	F	T	F	T
F	T	F	T	F	F	F
F	F	T	T	F	T	F
F	F	F	F	T	T	T



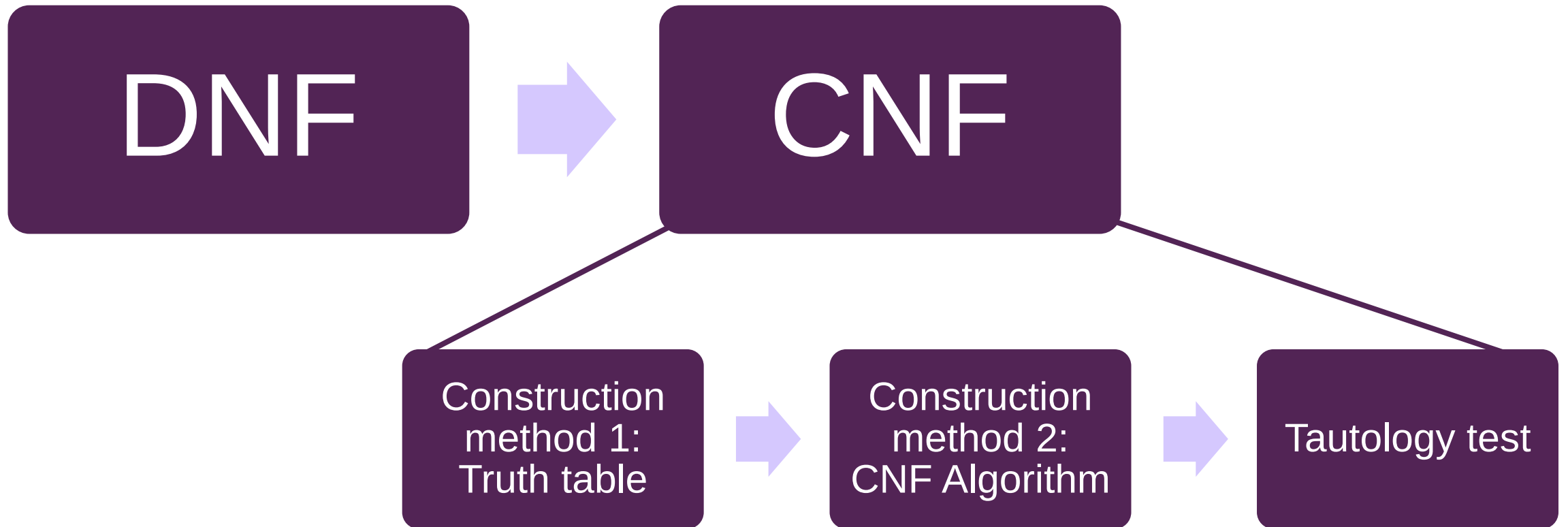
A, B and C are all liars

Planning Day 1



Break!

CNF + DNF



Disjunctive Normal Form (DNF)

- Formulas of the form:

$$(.. \wedge .. \wedge ..) \mathbf{V} (.. \wedge .. \wedge ..) \mathbf{V} (.. \wedge .. \wedge ..)$$

- Disjunction where the formulas are conjunctions of literals ($p/\neg p/q/\neg r/\dots$)

QUESTION:

Is $p \vee (p \wedge q)$ in DNF?

QUESTION:

Is $p \vee (p \wedge q)$ in DNF?



QUESTION:

Is $p \vee (p \wedge q)$ in DNF?



QUESTION 2:

Is $(q \rightarrow p) \vee r$ in DNF?

QUESTION:

Is $p \vee (p \wedge q)$ in DNF?



QUESTION 2:

Is $(q \rightarrow p) \vee r$ in DNF?



QUESTION:

Is $p \vee (p \wedge q)$ in DNF?



QUESTION 2:

Is $(q \rightarrow p) \vee r$ in DNF?



QUESTION 3:

Is p in DNF?

QUESTION:

Is $p \vee (p \wedge q)$ in DNF?



QUESTION 2:

Is $(q \rightarrow p) \vee r$ in DNF?



QUESTION 3:

Is p in DNF?



QUESTION:

Is $p \vee (p \wedge q)$ in DNF?



QUESTION 2:

Is $(q \rightarrow p) \vee r$ in DNF?



$(p \wedge q)$

QUESTION 3:

Is p in DNF?



QUESTION 4:

Is $p \wedge q$ in DNF?

QUESTION:

Is $p \vee (p \wedge q)$ in DNF?



QUESTION 2:

Is $(q \rightarrow p) \vee r$ in DNF?



QUESTION 3:

Is p in DNF?



QUESTION 4:

Is $p \wedge q$ in DNF?



Constructing DNF

p	q	r	?
T	T	T	F
T	T	F	F
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

Strategy



Step 1: Identify
all rows that
are true



Constructing DNF

p	q	r	?
T	T	T	F
T	T	F	F
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

Use negation
for False literals

$$p \wedge \neg q \wedge \neg r$$

Strategy

Step 1: Identify
all rows that
are true

Step 2:
Connect
literals using $\wedge$

Constructing DNF

p	q	r	?
T	T	T	F
T	T	F	F
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

Use negation
for False literals

$$p \wedge \neg q \wedge \neg r$$

$$\neg p \wedge q \wedge \neg r$$

Strategy

Step 1: Identify
all rows that
are true

Step 2:
Connect
literals using $\wedge$

Constructing DNF

p	q	r	?
T	T	T	F
T	T	F	F
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

Use negation
for False literals

$$p \wedge \neg q \wedge \neg r$$

$$\neg p \wedge q \wedge \neg r$$

$$\neg p \wedge \neg q \wedge \neg r$$

Strategy

Step 1: Identify
all rows that
are true

Step 2:
Connect
literals using $\wedge$

Constructing DNF

p	q	r	?
T	T	T	F
T	T	F	F
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T



$$p \wedge \neg q \wedge \neg r$$



$$\neg p \wedge q \wedge \neg r$$



$$\neg p \wedge \neg q \wedge \neg r$$

V

V

Strategy



Step 1: Identify
all rows that are
true



Step 2: Connect
literals using $\wedge$



Step 3: Connect
conjunctions
using $\vee$

Constructing DNF

p	q	r	?
T	T	T	F
T	T	F	F
T	F	T	F
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T



$(p \wedge \neg q \wedge \neg r)$



$(\neg p \wedge q \wedge \neg r)$



$(\neg p \wedge \neg q \wedge \neg r)$

$\vee$

$\vee$

$? \equiv$

Strategy



Step 1: Identify
all rows that are
true



Step 2: Connect
literals using $\wedge$



Step 3: Connect
conjunctions
using $\vee$

DNF: Strategy

DNF



Step 1: Identify
all rows that are
true



Step 2: Connect
literals using $\wedge$



Step 3: Connect
conjunctions
using $\vee$

Just a formula and
no truth table?
Construct it yourself!

Use negation
for False literals

This step creates
the disjunction we
want!

DNF: Exercise!

Exam Question: resit 2019

Disjunctive normal form (3+6 points)

- (a) Construct the truth table for the formula $(p \vee q) \rightarrow \neg r$.
- (b) Extract from this truth table a formula in DNF that is semantically equivalent to $(p \vee q) \rightarrow \neg r$.

p	q	r	$p \vee q$	$\neg r$	$(p \vee q) \rightarrow \neg r$	
T	T	T	T	F	F	
T	T	F	T	T	T	$\rightarrow (p \wedge q \wedge \neg r)$
T	F	T	T	F	F	
T	F	F	T	T	T	$\rightarrow (p \wedge \neg q \wedge \neg r)$
F	T	T	T	F	F	
F	T	F	T	T	T	$\rightarrow (\neg p \wedge q \wedge \neg r)$
F	F	T	F	F	T	$\rightarrow (\neg p \wedge \neg q \wedge r)$
F	F	F	F	T	T	$\rightarrow (\neg p \wedge \neg q \wedge \neg r)$

DNF



Step 1: Identify all rows that are true



Step 2: Connect literals using $\wedge$



Step 3: Connect conjunctions using $\vee$

DNF: Exercise!

Exam Question: resit 2019

Disjunctive normal form (3+6 points)

- (a) Construct the truth table for the formula $(p \vee q) \rightarrow \neg r$.
- (b) Extract from this truth table a formula in DNF that is semantically equivalent to $(p \vee q) \rightarrow \neg r$.

p	q	r	$\neg r$	$p \vee q$	$(p \vee q) \rightarrow \neg r$
T	T	T	F	T	F
T	T	F	T	T	T
T	F	T	F	T	F
T	F	F	T	T	T
F	T	T	F	T	F
F	T	F	T	T	T
F	F	T	F	F	T
F	F	F	T	F	T

$(p \wedge q \wedge \neg r)$

✓

$(p \wedge \neg q \wedge \neg r)$

✓

$(\neg p \wedge q \wedge \neg r)$

✓

$(\neg p \wedge \neg q \wedge r)$

✓

$(\neg p \wedge \neg q \wedge \neg r)$

DNF



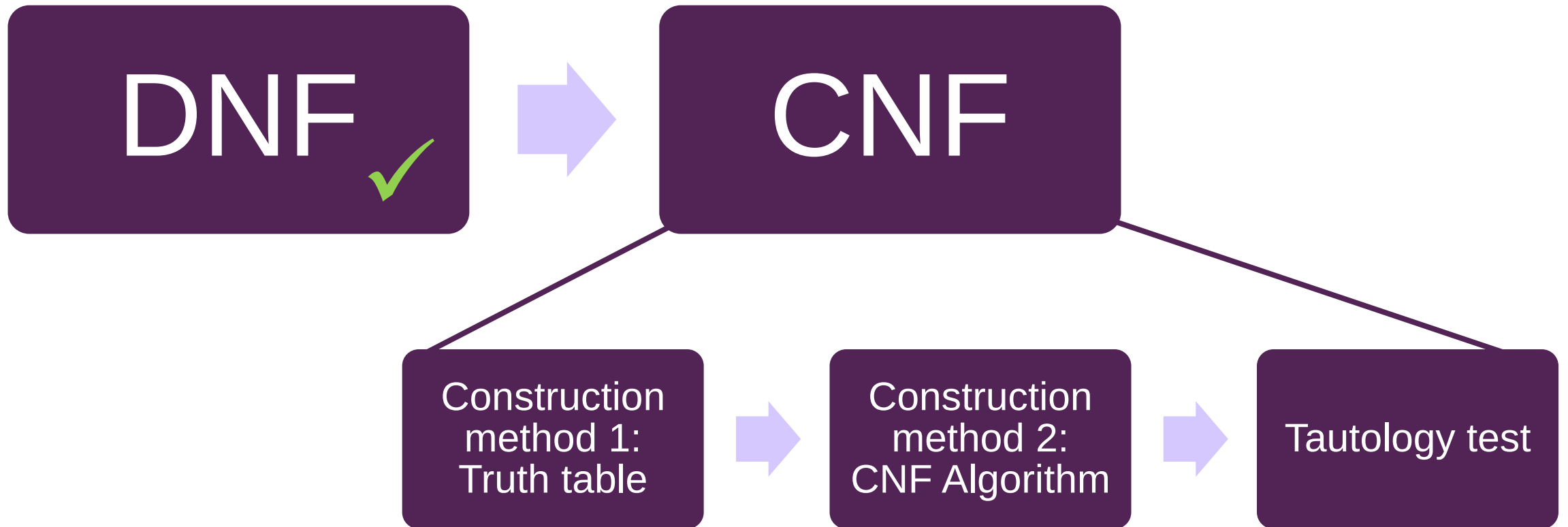
Step 1: Identify all rows that are true



Step 2: Connect literals using $\wedge$



Step 3: Connect conjunctions using $\vee$



Conjunctive Normal Form (CNF)

- Formulas of the form:

$$(.. \vee .. \vee ..) \mathbf{\wedge} (.. \vee .. \vee ..) \mathbf{\wedge} (.. \vee .. \vee ..)$$

- Conjunction where the formulas are disjunctions of literals ($p/\neg p/q/\neg r/\dots$)

Conjunctive Normal Form (CNF)

- Formulas of the form:

$$(.. \vee .. \vee ..) \wedge (.. \vee .. \vee ..) \wedge (.. \vee .. \vee ..)$$

- Conjunction where the formulas are disjunctions of literals ($p/\neg p/q/\neg r/\dots$)
 - Clause**: disjunction of literals
 - So, CNF = conjunctions of clauses

Constructing CNF 1

A CNF representing '?' should express that we are not in one of the three false rows

p	q	r	?
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

Strategy



Step 1: Identify all rows that are false



Constructing CNF 1

A CNF representing '?' should express that we are not in one of the three false rows

p	q	r	?
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

$$\neg p \vee \neg q \vee \neg r$$

Use negation for True literals

Strategy

Step 1: Identify all rows that are false

Step 2: Connect literals using $\vee$

Constructing CNF 1

A CNF representing '?' should express that we are not in one of the three false rows

p	q	r	?
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T



$$\neg p \vee \neg q \vee \neg r$$



$$p \vee \neg q \vee \neg r$$

Use negation
for True literals

Strategy



Step 1: Identify
all rows that are
false



Step 2: Connect
literals using $\vee$



Constructing CNF 1

A CNF representing '?' should express that we are not in one of the three false rows

p	q	r	?
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T



$$\neg p \vee \neg q \vee \neg r$$



$$p \vee \neg q \vee \neg r$$



$$p \vee q \vee \neg r$$

Use negation
for True literals

Strategy



Step 1: Identify
all rows that are
false



Step 2: Connect
literals using $\vee$



Constructing CNF 1

A CNF representing '?' should express that we are not in one of the three false rows

p	q	r	?
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

←

←

←

$\neg p \vee \neg q \vee \neg r$

$p \vee \neg q \vee \neg r$

$p \vee q \vee \neg r$

$\wedge$

$\wedge$

Strategy



Step 1: Identify all rows that are false



Step 2: Connect literals using $\vee$



Step 3: Connect disjunctions using $\wedge$

Constructing CNF 1

A CNF representing '?' should express that we are not in one of the three false rows

p	q	r	?
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	T
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	T

← $(\neg p \vee \neg q \vee \neg r)$

$\wedge$

← $(p \vee \neg q \vee \neg r)$

$\wedge$

← $(p \vee q \vee \neg r)$

? $\equiv$

Strategy



Step 1: Identify all rows that are false

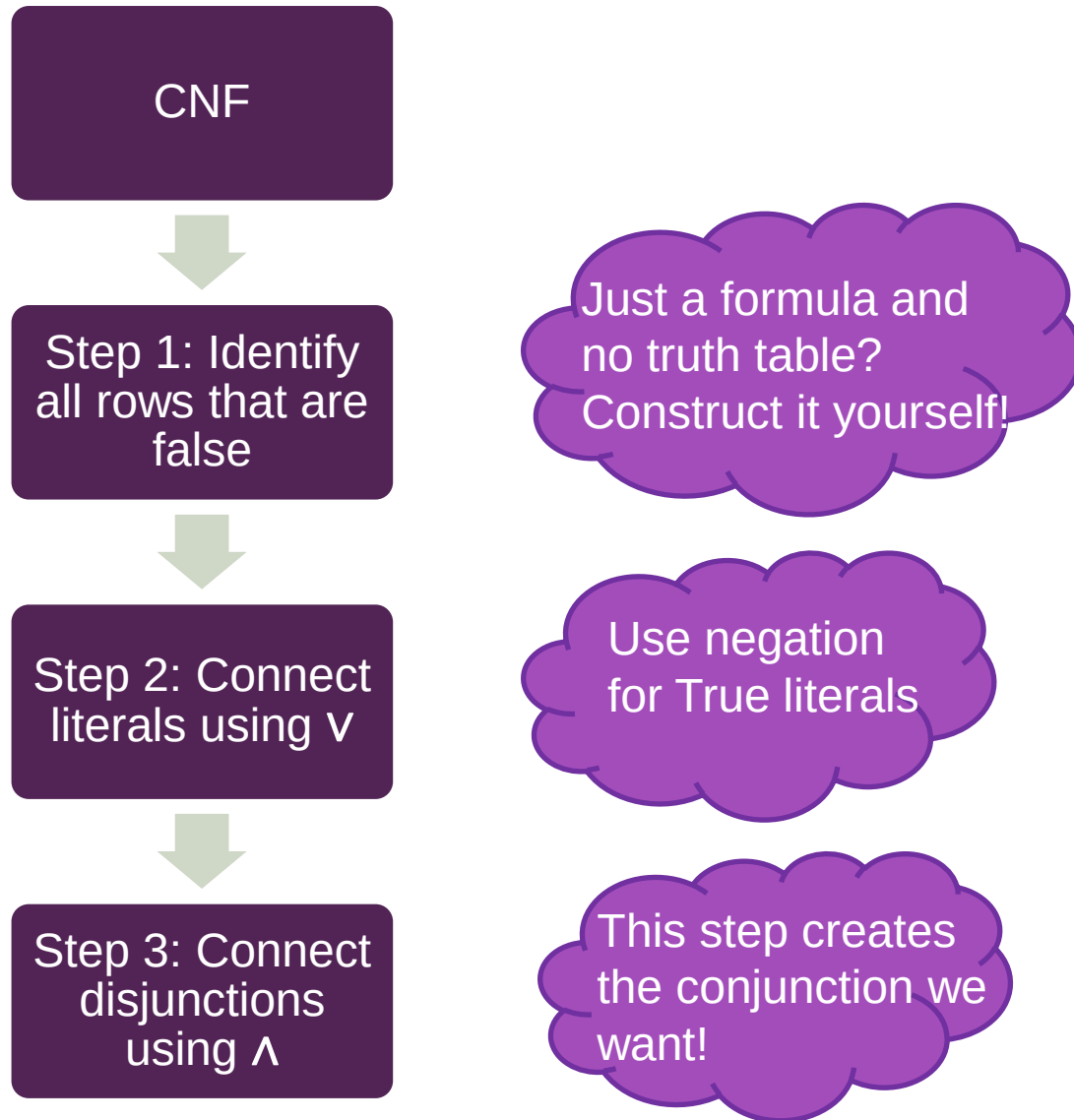


Step 2: Connect literals using $\vee$



Step 3: Connect disjunctions using $\wedge$

Strategy



CNF: Exercise!

Exam Question: midterm 2019

Conjunctive normal form (8 points)

Give the truth table of the DNF $(p \wedge q) \vee r$, and use it to construct a formula in CNF that is semantically equivalent.

p	q	r	$p \wedge q$	$(p \wedge q) \vee r$
T	T	T	T	T
T	T	F	T	T
T	F	T	F	T
T	F	F	F	F
F	T	T	F	T
F	T	F	F	F
F	F	T	F	T
F	F	F	F	F

$\nearrow$ CNF
 $\rightarrow (p \vee q \vee r)$
 $\rightarrow (p \vee q \vee r) \wedge (p \vee r \vee r)$
 $\rightarrow (p \vee q \vee r) \wedge (p \vee r)$

CNF



Step 1: Identify all rows that are false



Step 2: Connect literals using $\vee$



Step 3: Connect disjunctions using $\wedge$

CNF: Exercise!

Exam Question: midterm 2019

Conjunctive normal form (8 points)

Give the truth table of the DNF $(p \wedge q) \vee r$, and use it to construct a formula in CNF that is semantically equivalent.

p	q	r	$p \wedge q$	$(p \wedge q) \vee r$
T	T	T	T	T
T	T	F	T	T
T	F	T	F	T
T	F	F	F	F
F	T	T	F	T
F	T	F	F	F
F	F	T	F	T
F	F	F	F	F

$(\neg p \vee q \vee r)$

$\wedge$

$(p \vee \neg q \vee r)$

$\wedge$

$(p \vee q \vee r)$

CNF



Step 1: Identify all rows that are false



Step 2: Connect literals using $\vee$



Step 3: Connect disjunctions using $\wedge$

CNF: Exercise!

Exam Question: Final 2022

CNF (8 points)

Build a formula in CNF that corresponds to the following truth table (using the method explained in the third logic lecture).

p	q	r	$?$
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	F

CNF



Step 1: Identify
all rows that are
false



Step 2: Connect
literals using $\vee$



Step 3: Connect
disjunctions
using $\wedge$

CNF: Exercise!

Exam Question: Final 2022

CNF (8 points)

Build a formula in CNF that corresponds to the following truth table (using the method explained in the third logic lecture).

p	q	r	$?$
T	T	T	F
T	T	F	T
T	F	T	T
T	F	F	F
F	T	T	F
F	T	F	T
F	F	T	F
F	F	F	F

Solution: $(\neg p \vee \neg q \vee \neg r) \wedge (\neg p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge (p \vee q \vee \neg r) \wedge (p \vee q \vee r)$

CNF



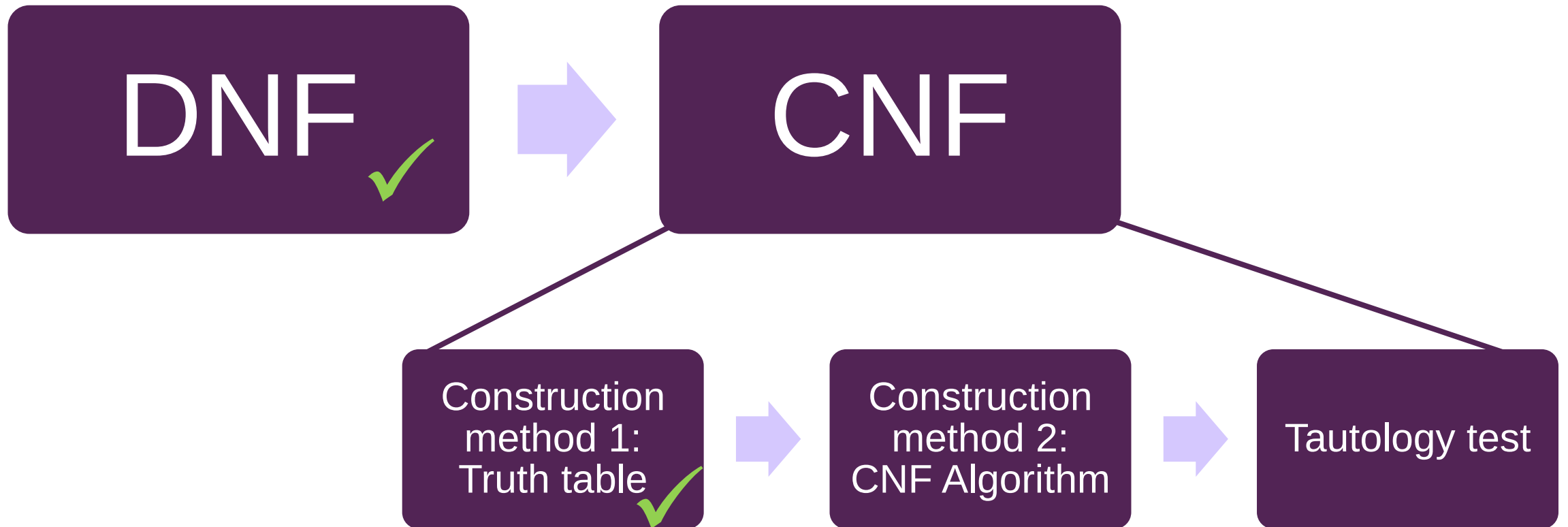
Step 1: Identify all rows that are false



Step 2: Connect literals using $\vee$



Step 3: Connect disjunctions using $\wedge$



CNF Algorithm: from formula to CNF

3 steps:

1. IMPL-FREE: eliminate all implications
2. NNF: Bring all negations inside the brackets
3. DISTR: re-arrange $\wedge$ and $\vee$ using distributivity rules

Step 1: Eliminate all implications

- $\Phi \rightarrow \Psi \equiv \neg \Phi \vee \Psi$

Constructing CNF 2

Step 1: Eliminate all implications

$$(\neg p \vee q) \rightarrow q$$

- $\Phi \rightarrow \Psi \equiv \neg \Phi \vee \Psi$

Constructing CNF 2

Step 1: Eliminate all implications

$$(\neg p \vee q) \rightarrow q$$

- $\Phi \rightarrow \Psi \equiv \neg \Phi \vee \Psi$

Constructing CNF 2

Step 1: Eliminate all implications

- $\Phi \rightarrow \Psi \equiv \neg \Phi \vee \Psi$

$$(\neg p \vee q) \rightarrow q$$



$$\neg(\neg p \vee q) \vee q$$

Constructing CNF 2

Step 2: Bring all negations inside the brackets

All negations have to be directly in front of a propositional variable.

- $\neg(\Phi \wedge \Psi) \equiv \neg\Phi \vee \neg\Psi$
- $\neg(\Phi \vee \Psi) \equiv \neg\Phi \wedge \neg\Psi$
- $\neg\neg\Phi \equiv \Phi$

$$(\neg p \vee q) \rightarrow q$$



$$\neg(\neg p \vee q) \vee q$$

Constructing CNF 2

Step 2: Bring all negations inside the brackets

All negations have to be directly in front of a propositional variable.

- $\neg(\Phi \wedge \Psi) \equiv \neg\Phi \vee \neg\Psi$
- $\neg(\Phi \vee \Psi) \equiv \neg\Phi \wedge \neg\Psi$
- $\neg\neg\Phi \equiv \Phi$

$$(\neg p \vee q) \rightarrow q$$



$$\neg(\neg p \vee q) \vee q$$

Constructing CNF 2

Step 2: Bring all negations inside the brackets

All negations have to be directly in front of a propositional variable.

- $\neg(\Phi \wedge \Psi) \equiv \neg\Phi \vee \neg\Psi$
- $\neg(\Phi \vee \Psi) \equiv \neg\Phi \wedge \neg\Psi$
- $\neg\neg\Phi \equiv \Phi$

$$(\neg p \vee q) \rightarrow q$$

$$\neg(\neg p \vee q) \vee q$$

$$(\neg\neg p \wedge \neg q) \vee q$$

Constructing CNF 2

Step 2: Bring all negations inside the brackets

All negations have to be directly in front of a propositional variable.

- $\neg(\Phi \wedge \Psi) \equiv \neg\Phi \vee \neg\Psi$
- $\neg(\Phi \vee \Psi) \equiv \neg\Phi \wedge \neg\Psi$
- $\neg\neg\Phi \equiv \Phi$

$$(\neg p \vee q) \rightarrow q$$

$$\neg(\neg p \vee q) \vee q$$

$$(\neg\neg p \wedge \neg q) \vee q$$

Constructing CNF 2

Step 2: Bring all negations inside the brackets

All negations have to be directly in front of a propositional variable.

- $\neg(\Phi \wedge \Psi) \equiv \neg\Phi \vee \neg\Psi$
- $\neg(\Phi \vee \Psi) \equiv \neg\Phi \wedge \neg\Psi$
- $\neg\neg\Phi \equiv \Phi$

$$(\neg p \vee q) \rightarrow q$$

$$\neg(\neg p \vee q) \vee q$$

$$(\neg\neg p \wedge \neg q) \vee q$$

$$(p \wedge \neg q) \vee q$$

Constructing CNF 2

Step 3: re-arrange $\wedge$ and $\vee$ using distributivity rules

- $(\Phi \wedge \Psi) \vee X \equiv (\Phi \vee X) \wedge (\Psi \vee X)$
- $X \vee (\Phi \wedge \Psi) \equiv (X \vee \Phi) \wedge (X \vee \Psi)$

$$(\neg p \vee q) \rightarrow q$$

$$\neg(\neg p \vee q) \vee q$$

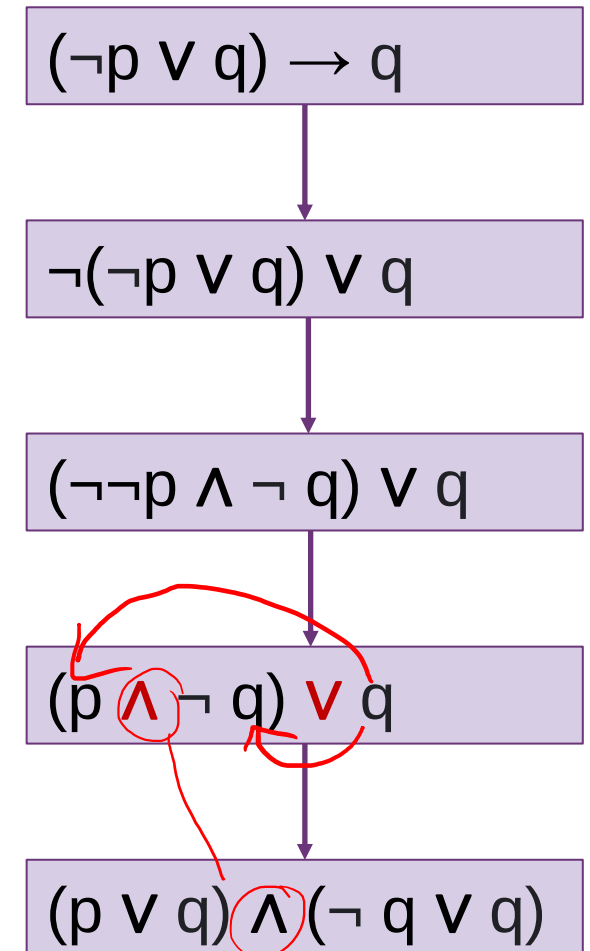
$$(\neg\neg p \wedge \neg q) \vee q$$

$$(p \wedge \neg q) \vee q$$

Constructing CNF 2

Step 3: re-arrange $\wedge$ and $\vee$ using distributivity rules

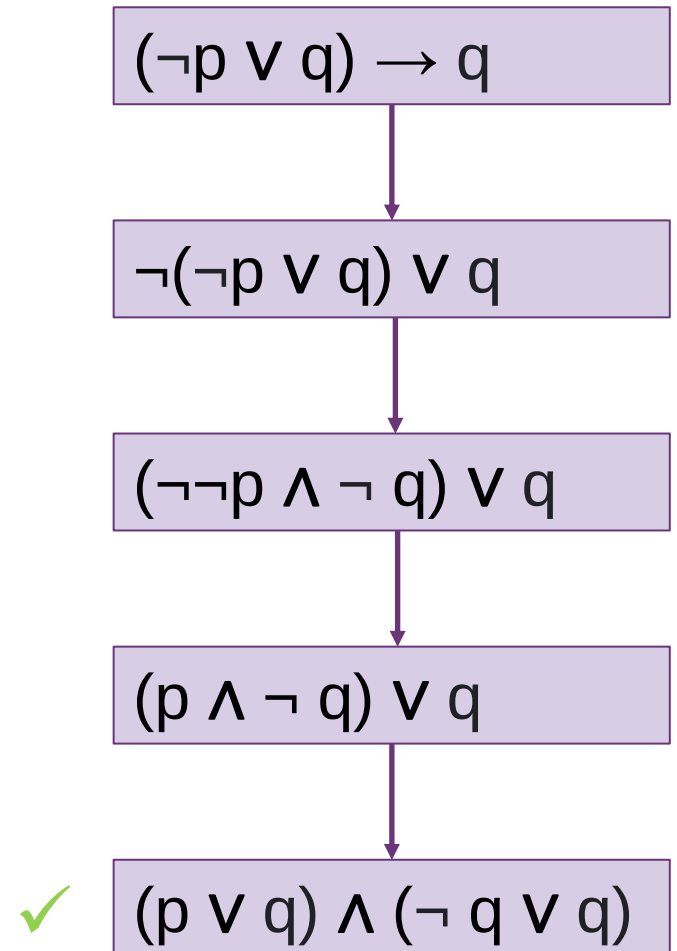
- $(\Phi \wedge \Psi) \vee X \equiv (\Phi \vee X) \wedge (\Psi \vee X)$
- $X \vee (\Phi \wedge \Psi) \equiv (X \vee \Phi) \wedge (X \vee \Psi)$



Constructing CNF 2

Step 3: re-arrange $\wedge$ and $\vee$ using distributivity rules

- $(\Phi \wedge \Psi) \vee X \equiv (\Phi \vee X) \wedge (\Psi \vee X)$
- $X \vee (\Phi \wedge \Psi) \equiv (X \vee \Phi) \wedge (X \vee \Psi)$



CNF: Exercise!

Exam Question: Midterm 2021

Conjunctive normal form (12 points)

Apply the algorithm CNF to turn the formula

$$\neg((p \vee q) \rightarrow r) \vee (\neg p \wedge r)$$

into CNF.

$$\neg((p \vee q) \rightarrow r) \vee (\neg p \wedge r)$$

$$\text{IMPL-FREE: } \neg(\neg(p \vee q) \vee r) \vee (\neg p \wedge r)$$

$$\text{NNF: } (\neg\neg(p \vee q) \wedge \neg r) \vee (\neg p \wedge r)$$

$$\text{NNF: } ((p \vee q) \wedge \neg r) \vee (\neg p \wedge r)$$

$$\text{DISTR: } ((p \vee q) \vee (\neg p \wedge r)) \wedge (\neg r \vee (\neg p \wedge r))$$

$$\text{DISTR: } ((p \vee q) \vee \neg p) \wedge ((p \vee q) \vee r) \wedge (\neg r \vee (\neg p)) \wedge (\neg r \vee r) \\ \rightarrow \text{CNF}$$

Step 1: IMPL-FREE

- $\Phi \rightarrow \Psi \equiv \neg \Phi \vee \Psi$

Step 2: NNF

- $\neg(\Phi \wedge \Psi) \equiv \neg \Phi \vee \neg \Psi$

- $\neg(\Phi \vee \Psi) \equiv \neg \Phi \wedge \neg \Psi$

- $\neg\neg\Phi \equiv \Phi$

Step 3: DISTR

- $(\Phi \wedge \Psi) \vee X \equiv (\Phi \vee X) \wedge (\Psi \vee X)$

- $X \vee (\Phi \wedge \Psi) \equiv (X \vee \Phi) \wedge (X \vee \Psi)$

CNF: Exercise!

Exam Question: Midterm 2021

Conjunctive normal form (12 points)

Apply the algorithm CNF to turn the formula

$$\neg((p \vee q) \rightarrow r) \vee (\neg p \wedge r)$$

into CNF.

Solution:

$$\neg((p \vee q) \rightarrow r) \vee (\neg p \wedge r)$$

$$1. \neg(\neg(p \vee q) \vee r) \vee (\neg p \wedge r)$$

$$2. (\neg\neg(p \vee q) \wedge \neg r) \vee (\neg p \wedge r)$$

$$3. ((p \vee q) \wedge \neg r) \vee (\neg p \wedge r)$$

$$4. ((p \vee q) \vee (\neg p \wedge r)) \wedge (\neg r \vee (\neg p \wedge r))$$

$$5. (((p \vee q) \vee \neg p) \wedge ((p \vee q) \vee r)) \wedge (\neg r \vee (\neg p \wedge r))$$

$$6. (((p \vee q) \vee \neg p) \wedge ((p \vee q) \vee r)) \wedge ((\neg r \vee \neg p) \wedge (\neg r \vee r))$$

IMPL-FREE

NNF

NNF

DISTR

DISTR

DISTR

Step 1: IMPL-FREE

$$\bullet \quad \Phi \rightarrow \Psi \equiv \neg \Phi \vee \Psi$$

Step 2: NNF

$$\bullet \quad \neg(\Phi \wedge \Psi) \equiv \neg \Phi \vee \neg \Psi$$

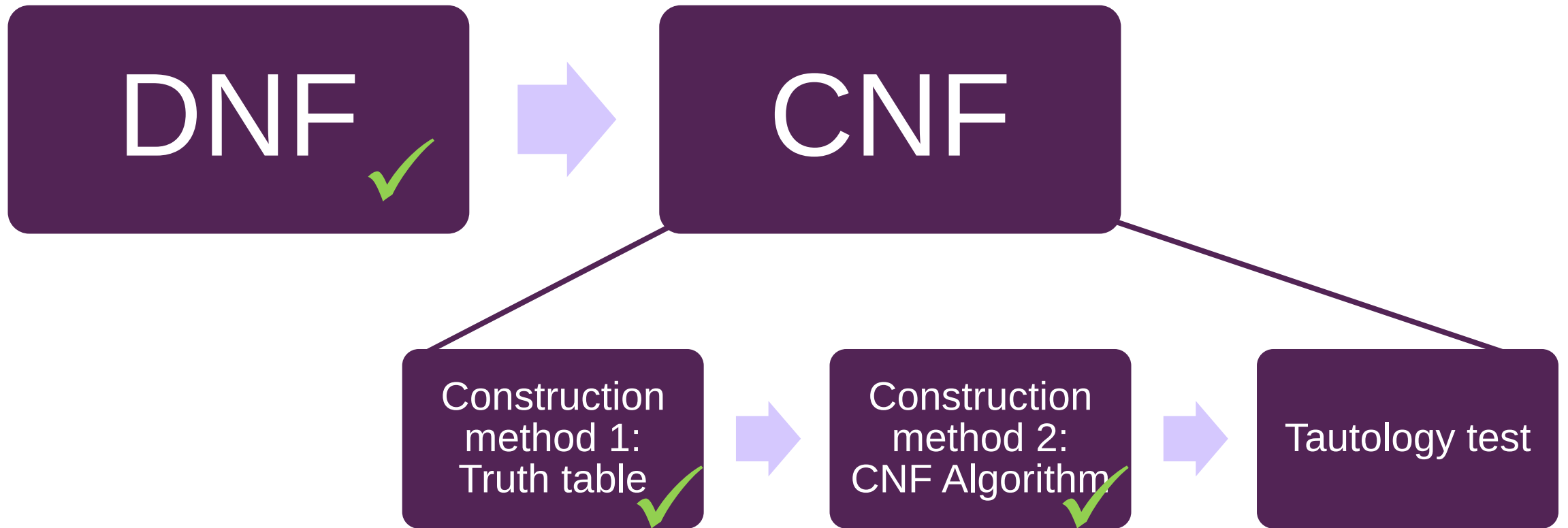
$$\bullet \quad \neg(\Phi \vee \Psi) \equiv \neg \Phi \wedge \neg \Psi$$

$$\bullet \quad \neg\neg \Phi \equiv \Phi$$

Step 3: DISTR

$$\bullet \quad (\Phi \wedge \Psi) \vee X \equiv (\Phi \vee X) \wedge (\Psi \vee X)$$

$$\bullet \quad X \vee (\Phi \wedge \Psi) \equiv (X \vee \Phi) \wedge (X \vee \Psi)$$



Tautology Test

How to see if a CNF is a tautology?

A CNF is a tautology if and only if **every** clause of the conjunction is a tautology as well

$$(.. \vee .. \vee ..) \wedge (.. \vee .. \vee ..) \wedge (.. \vee .. \vee ..)$$

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$$(.. \vee .. \vee ..) \wedge (.. \vee .. \vee ..) \wedge (.. \vee .. \vee ..)$$

A clause is a tautology if and only if it contains p and $\neg p$ for some p

P	$\neg P$	$P \vee \neg P$
T	F	T
F	T	T

$\begin{matrix} p \\ r \end{matrix}$
 $\begin{matrix} \neg p \\ \neg r \end{matrix}$

Tautology Test

How to see if a CNF is a tautology?

A CNF is a tautology if and only if **every** clause of the conjunction is a tautology as well

$$(.. \vee .. \vee ..) \wedge (.. \vee .. \vee ..) \wedge (.. \vee .. \vee ..)$$

A clause is a tautology if and only if it contains p and $\neg p$ for some p

e.g. $(q \vee r \vee p \vee \neg r)$

$\hookrightarrow$ tautology

Tautology Test

QUESTION:

Is $(q \vee r) \wedge \neg r$ a tautology?

Tautology Test

QUESTION:

Is $(q \vee r) \wedge \neg r$ a tautology?

x

Tautology Test

QUESTION:

Is $(\underline{p} \vee \neg q \vee \underline{\neg p}) \wedge (\neg p \vee \underline{\neg q} \vee \underline{q}) \wedge (\underline{\neg p} \vee \underline{p})$ a tautology?

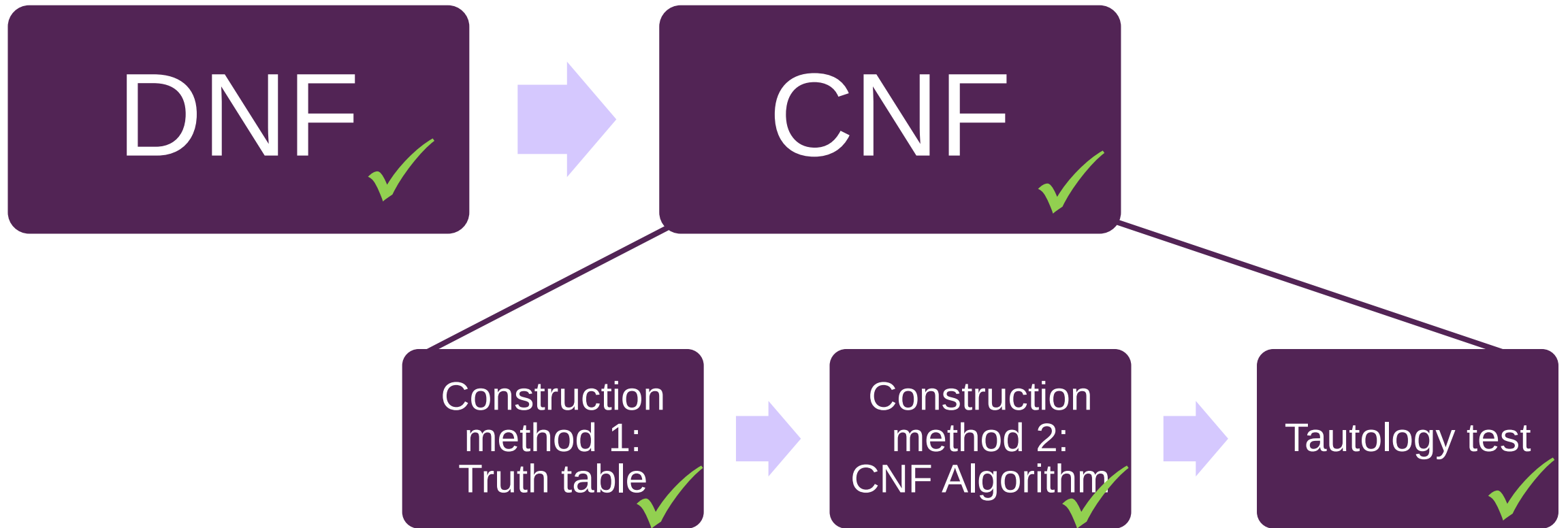
Tautology Test

QUESTION:

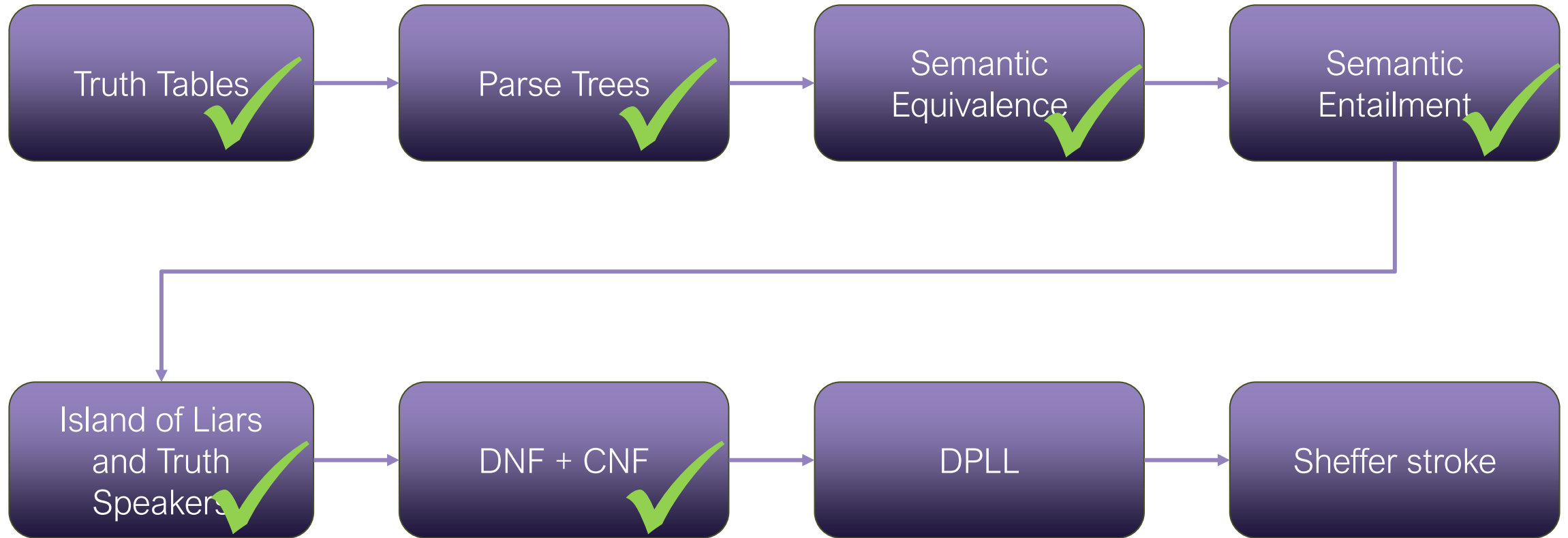
Is $(p \vee \neg q \vee \neg p) \wedge (\neg p \vee \neg q \vee q) \wedge (\neg p \vee p)$ a tautology?



Planning



Planning Day 1



DPLL

Satisfiability problem: Given a propositional formula ϕ , try to find a valuation that applied to ϕ yields $\top$.

$\longrightarrow P = \top, Q = F$ e.g.

DPLL checks if a CNF ϕ is satisfiable, and finds a valuation!

Atomic propositions $\top$ ('constant true') and $\perp$ ('constant false')

$\top$ and $\perp$ are *constants* (i.e., have no arguments), where:

- ▶ under every valuation, $\top$ has truth value T
- ▶ under every valuation, $\perp$ has truth value F

The DPLL procedure applies the following reduction rules to eliminate connectives with $\top$ or $\perp$:

$$\psi \wedge \top \rightsquigarrow \psi$$

$$\top \wedge \psi \rightsquigarrow \psi$$

$$\psi \wedge \perp \rightsquigarrow \perp$$

$$\perp \wedge \psi \rightsquigarrow \perp$$

$$\psi \vee \top \rightsquigarrow \top$$

$$\top \vee \psi \rightsquigarrow \top$$

$$\psi \vee \perp \rightsquigarrow \psi$$

$$\perp \vee \psi \rightsquigarrow \psi$$

$$\neg \top \rightsquigarrow \perp$$

$$\neg \perp \rightsquigarrow \top$$

DPLL: Step Plan

- ▶ eliminate “unit” clauses: $\rightarrow$ single literal
 - ▶ for each *unit clause* p of ϕ , replace all p 's in ϕ by $\top$
 - ▶ for each *unit clause* $\neg p$ of ϕ , replace all p 's in ϕ by $\perp$ $\rightarrow F = \top$
- ▶ eliminate “pure” propositional variables:
 - ▶ if a p occurs only *positively* in ϕ , replace all p 's in ϕ by $\top$
 - ▶ if a p occurs only *negatively* in ϕ , replace all p 's in ϕ by $\perp$
- ▶ if ϕ becomes $\top$, then it is *satisfiable*
- ▶ if ϕ becomes $\perp$, then it is *unsatisfiable*
- ▶ choose a p in ϕ :
 - ▶ replace all p 's in ϕ by $\top$, and apply the DPLL procedure
 - ▶ if the outcome is “unsatisfiable”, then replace all p 's in ϕ by $\perp$, and *again* apply the DPLL procedure

DPLL: Example

Exam Question: midterm 2020

4. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF $(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$, to check whether it is satisfiable. Explain in detail how the procedure comes to a result.

$$(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$$

$p = F$

▶ eliminate “unit” clauses:

- ▶ for each *unit clause* p of ϕ , replace all p 's in ϕ by $\top$
- ▶ for each *unit clause* $\neg p$ of ϕ , replace all p 's in ϕ by $\perp$

▶ eliminate “pure” propositional variables:

- ▶ if a p occurs only *positively* in ϕ , replace all p 's in ϕ by $\top$
- ▶ if a p occurs only *negatively* in ϕ , replace all p 's in ϕ by $\perp$

▶ if ϕ becomes $\top$, then it is *satisfiable*

▶ if ϕ becomes $\perp$, then it is *unsatisfiable*

▶ choose a p in ϕ :

- ▶ replace all p 's in ϕ by $\top$, and apply the DPLL procedure
- ▶ if the outcome is “unsatisfiable”, then replace all p 's in ϕ by $\perp$, and *again* apply the DPLL procedure

$\psi \wedge \top \rightsquigarrow \psi$	$\psi \vee \top \rightsquigarrow \top$	$\neg \top \rightsquigarrow \perp$
$\top \wedge \psi \rightsquigarrow \psi$	$\top \vee \psi \rightsquigarrow \top$	$\neg \perp \rightsquigarrow \top$
$\psi \wedge \perp \rightsquigarrow \perp$	$\psi \vee \perp \rightsquigarrow \psi$	
$\perp \wedge \psi \rightsquigarrow \perp$	$\perp \vee \psi \rightsquigarrow \psi$	

DPLL: Example

Exam Question: midterm 2020

4. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF $(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$, to check whether it is satisfiable. Explain in detail how the procedure comes to a result.

$$(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$$

1. Remove unit clause:

$$(\perp \vee q \vee r) \wedge (\perp \vee \neg q \vee \neg r) \wedge \neg \perp$$

$$(q \vee r) \wedge (\neg q \vee \neg r) \wedge \top$$

$$(q \vee r) \wedge (\neg q \vee \neg r)$$

► eliminate “unit” clauses:

- for each *unit clause* p of ϕ , replace all p 's in ϕ by $\top$
- for each *unit clause* $\neg p$ of ϕ , replace all p 's in ϕ by $\perp$

► eliminate “pure” propositional variables:

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$$\psi \wedge \top \rightsquigarrow \psi$$

$$\top \wedge \psi \rightsquigarrow \psi$$

$$\psi \wedge \perp \rightsquigarrow \perp$$

$$\perp \wedge \psi \rightsquigarrow \perp$$

$$\psi \vee \top \rightsquigarrow \top$$

$$\top \vee \psi \rightsquigarrow \top$$

$$\psi \vee \perp \rightsquigarrow \psi$$

$$\perp \vee \psi \rightsquigarrow \psi$$

$$\neg \top \rightsquigarrow \perp$$

$$\neg \perp \rightsquigarrow \top$$

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Apply the DPLL procedure to the CNF $(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$, to check whether it is satisfiable. Explain in detail how the procedure comes to a result.

$$(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$$

1. Remove unit clause:

$$(\perp \vee q \vee r) \wedge (\perp \vee \neg q \vee \neg r) \wedge \neg \perp$$

$$(q \vee r) \wedge (\neg q \vee \neg r) \wedge \top$$

$$(q \vee r) \wedge (\neg q \vee \neg r)$$

2. Eliminate pure clauses: we don't have any!

► eliminate “unit” clauses:

- for each *unit clause* p of ϕ , replace all p 's in ϕ by $\top$
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$$\psi \wedge \top \rightsquigarrow \psi$$

$$\top \wedge \psi \rightsquigarrow \psi$$

$$\psi \wedge \perp \rightsquigarrow \perp$$

$$\perp \wedge \psi \rightsquigarrow \perp$$

$$\psi \vee \top \rightsquigarrow \top$$

$$\top \vee \psi \rightsquigarrow \top$$

$$\psi \vee \perp \rightsquigarrow \psi$$

$$\perp \vee \psi \rightsquigarrow \psi$$

$$\neg \top \rightsquigarrow \perp$$

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DPLL: Example

Exam Question: midterm 2020

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Apply the DPLL procedure to the CNF $(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$, to check whether it is satisfiable. Explain in detail how the procedure comes to a result.

$$(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$$

1. Remove unit clause:

$$(\perp \vee q \vee r) \wedge (\perp \vee \neg q \vee \neg r) \wedge \neg \perp$$

$$(q \vee r) \wedge (\neg q \vee \neg r) \wedge \top$$

$$(q \vee r) \wedge (\neg q \vee \neg r)$$

2. Eliminate pure clauses: we don't have any!

3. Choose a p: Let's pick q

$$(q \vee r) \wedge (\neg q \vee \neg r)$$

$$(T \vee r) \wedge (\neg T \vee \neg r)$$

$$\top \wedge (\perp \vee \neg r)$$

$$\neg r$$

► eliminate “unit” clauses:

- for each *unit clause* p of ϕ , replace all p 's in ϕ by $\top$
- for each *unit clause* $\neg p$ of ϕ , replace all p 's in ϕ by $\perp$

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$\psi \wedge \top \rightsquigarrow \psi$	$\psi \vee \top \rightsquigarrow \top$	$\neg \top \rightsquigarrow \perp$
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$\psi \wedge \perp \rightsquigarrow \perp$	$\psi \vee \perp \rightsquigarrow \psi$	
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$$(p \vee q \vee r) \wedge (p \vee \neg q \vee \neg r) \wedge \neg p$$

1. Remove unit clause:

$$(\perp \vee q \vee r) \wedge (\perp \vee \neg q \vee \neg r) \wedge \neg \perp$$

$$(q \vee r) \wedge (\neg q \vee \neg r) \wedge \top$$

$$(q \vee r) \wedge (\neg q \vee \neg r)$$

2. Eliminate pure clauses:

we don't have any!

3. Choose a p: Let's pick q

$$(q \vee r) \wedge (\neg q \vee \neg r)$$

$$(T \vee r) \wedge (\neg T \vee \neg r)$$

$$\top \wedge (\perp \vee \neg r)$$

$$\neg r$$

4. We only have $\neg r$ left, so assign $r \perp$

Then we have

$$\neg \perp \longrightarrow \top$$

So the formula is satisfiable! With values p is $\perp$, q is $\top$, r is $\perp$

$$\begin{aligned} p &= \perp \\ q &= \top \\ r &= \perp \end{aligned}$$

► eliminate "unit" clauses:

► for each *unit clause* p of ϕ , replace all p 's in ϕ by $\top$

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$\psi \wedge \perp \rightsquigarrow \perp$	$\psi \vee \perp \rightsquigarrow \psi$	
$\perp \wedge \psi \rightsquigarrow \perp$	$\perp \vee \psi \rightsquigarrow \psi$	

DPLL: Exercise!

Exam Question: resit 2021

2. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF

$$(\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee \neg r) \wedge (p \vee \neg q \vee \neg r) \wedge r$$

to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Let the procedure branch first on p , then on q , and then on r ; and let it always try the truth value true ($\top$) before false ($\perp$).

$$(\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee \neg r) \wedge (p \vee \neg q \vee \neg r) \wedge r$$

$$r = \top$$

$$(\neg p \vee q \vee \neg \top) \wedge (\neg p \vee \neg q \vee \neg \top) \wedge (p \vee \neg q \vee \neg \top)$$

$$(\neg p \vee q) \wedge (\neg p \vee \neg q) \wedge (p \vee \neg q)$$

$$p = \top$$

$$(\neg \top \vee q) \wedge (\neg \top \vee \neg q) \Rightarrow q \wedge \neg q \rightarrow \text{contradiction} \rightarrow \text{unsatisfiable}$$

$$p = \perp$$

$$(\top \vee q) \wedge (\top \vee \neg q) \wedge (\perp \vee \neg q)$$

$$q = \perp \Rightarrow \text{satisfiable when } r = \top, p = \perp, q = \perp$$

eliminate "unit" clauses:

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- for each *unit clause* $\neg p$ of ϕ , replace all p 's in ϕ by $\perp$

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$\top \wedge \psi \rightsquigarrow \psi$	$\top \vee \psi \rightsquigarrow \top$	$\neg \perp \rightsquigarrow \top$
$\psi \wedge \perp \rightsquigarrow \perp$	$\psi \vee \perp \rightsquigarrow \psi$	
$\perp \wedge \psi \rightsquigarrow \perp$	$\perp \vee \psi \rightsquigarrow \psi$	

DPLL: Exercise!

Exam Question: resit 2021

2. DPLL procedure (12 points)

Apply the DPLL procedure to the CNF

$$(\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee \neg r) \wedge (p \vee \neg q \vee \neg r) \wedge r$$

to check whether it is satisfiable. Explain in detail how the procedure comes to a result. Let the procedure branch first on p , then on q , and then on r ; and let it always try the truth value true ($\top$) before false ($\perp$).

$$(\neg p \vee q \vee \neg r) \wedge (\neg p \vee \neg q \vee \neg r) \wedge (p \vee \neg q \vee \neg r) \wedge r$$

Solution: The conjunct r is a unit clause, so substitute $\top$ for it:

$$(\neg p \vee q \vee \neg \top) \wedge (\neg p \vee \neg q \vee \neg \top) \wedge (p \vee \neg q \vee \neg \top) \wedge \top$$

After replacing the two occurrences of $\neg \top$ by $\perp$, this reduces to

$$(\neg p \vee q) \wedge (\neg p \vee \neg q) \wedge (p \vee \neg q)$$

This CNF formula contains no occurrences of unit clauses or pure variables.

Substituting $\top$ for p , replacing the two occurrences of $\neg \top$ by $\perp$, and reducing the formula yields $q \wedge \neg q \wedge \top$, which reduces further to $q \wedge \neg q$. After substituting $\top$ for q the formula reduces to $\perp$. Now backtrack and substitute $\perp$ for q . Again the formula reduces to $\perp$.

Now backtrack and substitute $\perp$ (instead of $\top$) for p . Replacing the two occurrences of $\neg \perp$ by $\top$ and reducing the formula yields $\top \wedge \top \wedge \neg q$, which reduces further to $\neg q$. After substituting $\top$ for q , the formula reduces to $\perp$. Now backtrack and substitute $\perp$ for q , in which case the formula reduces to $\top$.

► eliminate “unit” clauses:

- for each *unit clause* p of ϕ , replace all p 's in ϕ by $\top$
- for each *unit clause* $\neg p$ of ϕ , replace all p 's in ϕ by $\perp$

► eliminate “pure” propositional variables:

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► choose a p in ϕ :

- replace all p 's in ϕ by $\top$, and apply the DPLL procedure
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$$\psi \wedge \top \rightsquigarrow \psi$$

$$\top \wedge \psi \rightsquigarrow \psi$$

$$\psi \wedge \perp \rightsquigarrow \perp$$

$$\perp \wedge \psi \rightsquigarrow \perp$$

$$\psi \vee \top \rightsquigarrow \top$$

$$\top \vee \psi \rightsquigarrow \top$$

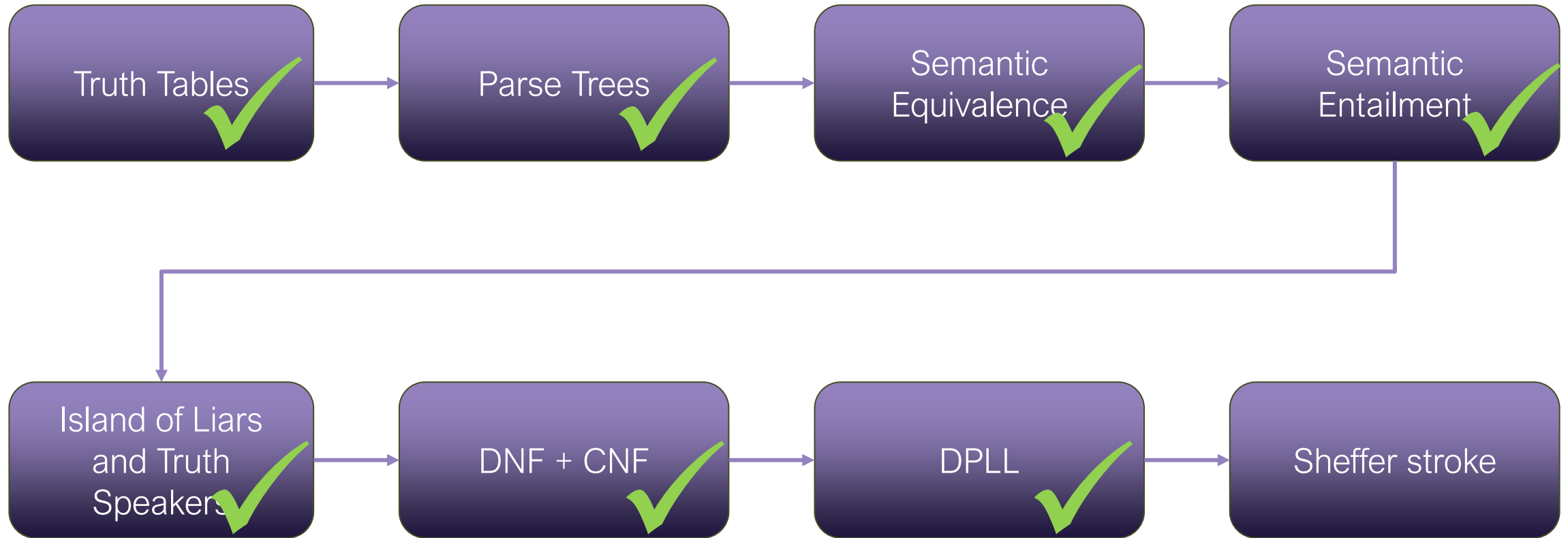
$$\psi \vee \perp \rightsquigarrow \psi$$

$$\perp \vee \psi \rightsquigarrow \psi$$

$$\neg \top \rightsquigarrow \perp$$

$$\neg \perp \rightsquigarrow \top$$

Planning Day 1



Sheffer Stroke

Sheffer Stroke

Definition:

$$\phi \mid \psi \equiv \neg(\phi \wedge \psi)$$

Sheffer Stroke

Definition:

$$\phi \mid \psi \equiv \neg(\phi \wedge \psi)$$

True as long as not both ϕ and ψ are true!
“NAND-operation” (= not and)

Truth table of $\phi \mid \psi$

ϕ	ψ	$\phi \mid \psi$
T	T	F
T	F	T
F	T	T
F	F	T

Sheffer Stroke

Question

What does $\phi \mid \phi$ mean?

$$\phi \mid \phi \equiv \neg \phi$$

Definition:

$$\phi \mid \psi \equiv \neg(\phi \wedge \psi)$$

$$\phi \mid \phi \equiv \neg(\phi \wedge \phi) \equiv \neg \phi \vee \neg \phi \equiv \neg \phi$$

Sheffer Stroke

Question

What does $\phi \mid \phi$ mean ?

Answer: $\neg\phi$

Definition:

$$\phi \mid \psi \equiv \neg(\phi \wedge \psi)$$

Sheffer Stroke

Question

What does $\phi \mid \phi$ mean ?

Answer: $\neg\phi$

Question

How can $\phi \vee \psi$ be expressed using only $\mid$ and $\neg$?

Hint: $\phi \vee \psi \equiv \neg(\neg\phi \wedge \neg\psi)$

$$\phi \vee \psi \equiv \neg(\neg\phi \wedge \neg\psi) \equiv \neg\phi \mid \neg\psi$$

Definition:

$$\phi \mid \psi \equiv \neg(\phi \wedge \psi)$$


Sheffer Stroke

Question

What does $\phi \mid \phi$ mean ?

Answer: $\neg\phi$

Question

How can $\phi \vee \psi$ be expressed using only $\mid$ and $\neg$?

Hint: $\phi \vee \psi \equiv \neg(\neg\phi \wedge \neg\psi)$

Answer: $\neg\phi \mid \neg\psi$

Definition:

$$\phi \mid \psi \equiv \neg(\phi \wedge \psi)$$

Sheffer Stroke

Question

What does $\phi | \phi$ mean?

Answer: $\neg\phi$

Question

How can $\phi \vee \psi$ be expressed using only $|$ and $\neg$?

Hint: $\phi \vee \psi \equiv \neg(\neg\phi \wedge \neg\psi)$

Answer: $\neg\phi | \neg\psi$

Question

How can $\phi \vee \psi$ be expressed using only $|$?

Definition:

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg\phi | \neg\psi \equiv (\phi | \phi) | (\psi | \psi)$$

Sheffer Stroke

Question

What does $\phi \mid \phi$ mean?

Answer: $\neg\phi$

Question

How can $\phi \vee \psi$ be expressed using only $\mid$ and $\neg$?

Hint: $\phi \vee \psi \equiv \neg(\neg\phi \wedge \neg\psi)$

Answer: $\neg\phi \mid \neg\psi$

Question

How can $\phi \vee \psi$ be expressed using only $\mid$?

Answer: $(\phi \mid \phi) \mid (\psi \mid \psi)$

$$\neg X \equiv X \mid X$$

Definition:

$$\phi \mid \psi \equiv \neg(\phi \wedge \psi)$$

Sheffer Stroke

Question

What does $\phi \mid \phi$ mean ?

Answer: $\neg\phi$

Question

How can $\phi \vee \psi$ be expressed using only $\mid$ and $\neg$?

Hint: $\phi \vee \psi \equiv \neg(\neg\phi \wedge \neg\psi)$

Answer: $\neg\phi \mid \neg\psi$

Question

How can $\phi \vee \psi$ be expressed using only $\mid$?

Answer: $(\phi \mid \phi) \mid (\psi \mid \psi)$

Definition:

$$\phi \mid \psi \equiv \neg(\phi \wedge \psi)$$

Tip: Try to remember these for the exam!

Sheffer Stroke: Example

Exam Question: midterm 2020

Sheffer stroke (12 points)

Express the propositional formula $\neg(\phi \oplus \psi)$ using only the Sheffer stroke $|$.

(Hint: Use the semantic equivalence $\phi \oplus \psi \equiv \neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi)$.)

Definition:

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg\phi \equiv \phi | \phi$$

$$\phi \vee \psi \equiv (\phi | \phi) | (\psi | \psi)$$

$$\neg(\phi \oplus \psi) \equiv \neg(\neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi))$$

Sheffer Stroke: Example

Exam Question: midterm 2020

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Express the propositional formula $\neg(\phi \oplus \psi)$ using only the Sheffer stroke $|$.

(Hint: Use the semantic equivalence $\phi \oplus \psi \equiv \neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi)$.)

Definition:

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg\phi \equiv \phi | \phi$$

$$\phi \vee \psi \equiv (\phi | \phi) | (\psi | \psi)$$

$$\begin{aligned} \neg(\phi \oplus \psi) &\equiv \neg(\neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi)) \\ &\equiv \neg((\phi | \psi) \wedge \neg((\phi | \phi) \wedge (\psi | \psi))) \end{aligned}$$

Sheffer Stroke: Example

Exam Question: midterm 2020

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Express the propositional formula $\neg(\phi \oplus \psi)$ using only the Sheffer stroke $|$.

(Hint: Use the semantic equivalence $\phi \oplus \psi \equiv \neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi)$.)

Definition:

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg\phi \equiv \phi | \phi$$

$$\phi \vee \psi \equiv (\phi | \phi) | (\psi | \psi)$$

$$\neg(\phi \oplus \psi) \equiv \neg(\neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi))$$

$$\equiv \neg((\phi | \psi) \wedge \neg((\phi | \phi) \wedge (\psi | \psi)))$$

$$\equiv \neg((\phi | \psi) \wedge ((\phi | \phi) | (\psi | \psi)))$$

Sheffer Stroke: Example

Exam Question: midterm 2020

Sheffer stroke (12 points)

Express the propositional formula $\neg(\phi \oplus \psi)$ using only the Sheffer stroke $|$.

(Hint: Use the semantic equivalence $\phi \oplus \psi \equiv \neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi)$.)

Definition:

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg\phi \equiv \phi | \phi$$

$$\phi \vee \psi \equiv (\phi | \phi) | (\psi | \psi)$$

$$\neg(\phi \oplus \psi) \equiv \neg(\neg(\phi \wedge \psi) \wedge \neg(\neg\phi \wedge \neg\psi))$$

$$\equiv \neg((\phi | \psi) \wedge \neg((\phi | \phi) \wedge (\psi | \psi)))$$

$$\equiv \neg(\underbrace{(\phi | \psi)}_{\phi} \wedge \underbrace{\neg((\phi | \phi) \wedge (\psi | \psi))}_{\psi}) \equiv \phi | \psi$$

$$\equiv \underbrace{(\phi | \psi)}_{\phi} | \underbrace{((\phi | \phi) | (\psi | \psi))}_{\psi}$$

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg(\phi \wedge \psi) \equiv \phi | \psi$$

Sheffer Stroke: Exercise!

Exam Question: resit 2021

Sheffer stroke (10 points)

Is the Sheffer stroke $|$ associative? If so, prove this via a truth table. If not, give a counterexample.

Definition:

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg\phi \equiv \phi | \phi$$

$$\phi \vee \psi \equiv (\phi | \phi) | (\psi | \psi)$$

Reminder:

Associativity:

$$\phi \vee (\psi \vee \chi) \equiv (\phi \vee \psi) \vee \chi$$

$$\phi \wedge (\psi \wedge \chi) \equiv (\phi \wedge \psi) \wedge \chi$$

What would associativity for $|$ look like?

Sheffer Stroke: Exercise!

Exam Question: resit 2021

Sheffer stroke (10 points)

Is the Sheffer stroke $|$ associative? If so, prove this via a truth table. If not, give a counterexample.

Definition:

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$$\neg\phi \equiv \phi | \phi$$

$$\phi \vee \psi \equiv (\phi | \phi) | (\psi | \psi)$$

Reminder:

Associativity:

$$\phi \vee (\psi \vee \chi) \equiv (\phi \vee \psi) \vee \chi$$

$$\phi \wedge (\psi \wedge \chi) \equiv (\phi \wedge \psi) \wedge \chi$$

What would associativity for $|$ look like?

ϕ	ψ	χ				

$$(\phi | \psi) | \chi \equiv \phi | (\psi | \chi)$$

Sheffer Stroke: Exercise!

Exam Question: resit 2021

Sheffer stroke (10 points)

Is the Sheffer stroke $|$ associative? If so, prove this via a truth table. If not, give a counterexample.

Definition:

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg\phi \equiv \phi | \phi$$

$$\phi \vee \psi \equiv (\phi | \phi) | (\psi | \psi)$$

Reminder:

Associativity:

$$\phi \vee (\psi \vee \chi) \equiv (\phi \vee \psi) \vee \chi$$

$$\phi \wedge (\psi \wedge \chi) \equiv (\phi \wedge \psi) \wedge \chi$$

What would associativity for $|$ look like?

ϕ	ψ	χ	$(\phi \psi)$	$(\phi \psi) \chi$	$(\psi \chi)$	$\phi (\psi \chi)$
T	T	T	F	T	F	T
T	T	F	F	T	T	F
T	F	T	T	F	T	F
T	F	F	T	T	T	F
F	T	T	T	F	F	T
F	T	F	T	T	T	T
F	F	T	T	F	T	T
F	F	F	T	T	T	T

counterexample:

$$\phi = T$$

$$\psi = T$$

$$\chi = F$$

Sheffer Stroke: Exercise!

Exam Question: resit 2021

Sheffer stroke (10 points)

Is the Sheffer stroke $|$ associative? If so, prove this via a truth table. If not, give a counterexample.

ϕ	ψ	χ	$(\phi \psi)$	$(\phi \psi) \chi$	$(\psi \chi)$	$\phi (\psi \chi)$
T	T	T	F	T	F	T
T	T	F	F	T	T	F
T	F	T	T	F	T	F
T	F	F	T	T	T	F
F	T	T	T	F	F	T
F	T	F	T	T	T	T
F	F	T	T	F	T	T
F	F	F	T	T	T	T

Definition:

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg\phi \equiv \phi | \phi$$

$$\phi \vee \psi \equiv (\phi | \phi) | (\psi | \psi)$$

Reminder:

Associativity:

$$\phi \vee (\psi \vee \chi) \equiv (\phi \vee \psi) \vee \chi$$

$$\phi \wedge (\psi \wedge \chi) \equiv (\phi \wedge \psi) \wedge \chi$$

What would associativity for $|$ look like?

Sheffer Stroke: Exercise!

Exam Question: resit 2021

Sheffer stroke (10 points)

Is the Sheffer stroke $|$ associative? If so, prove this via a truth table. If not, give a counterexample.

ϕ	ψ	χ	$(\phi \psi)$	$(\phi \psi) \chi$	$(\psi \chi)$	$\phi (\psi \chi)$
T	T	T	F	T	F	T
T	T	F	F	T	T	F
T	F	T	T	F	T	F
T	F	F	T	T	T	F
F	T	T	T	F	F	T
F	T	F	T	T	T	T
F	F	T	T	F	T	T
F	F	F	T	T	T	T

Definition:

$$\phi | \psi \equiv \neg(\phi \wedge \psi)$$

$$\neg\phi \equiv \phi | \phi$$

$$\phi \vee \psi \equiv (\phi | \phi) | (\psi | \psi)$$

Reminder:

Associativity:

$$\phi \vee (\psi \vee \chi) \equiv (\phi \vee \psi) \vee \chi$$

$$\phi \wedge (\psi \wedge \chi) \equiv (\phi \wedge \psi) \wedge \chi$$

What would associativity for $|$ look like?

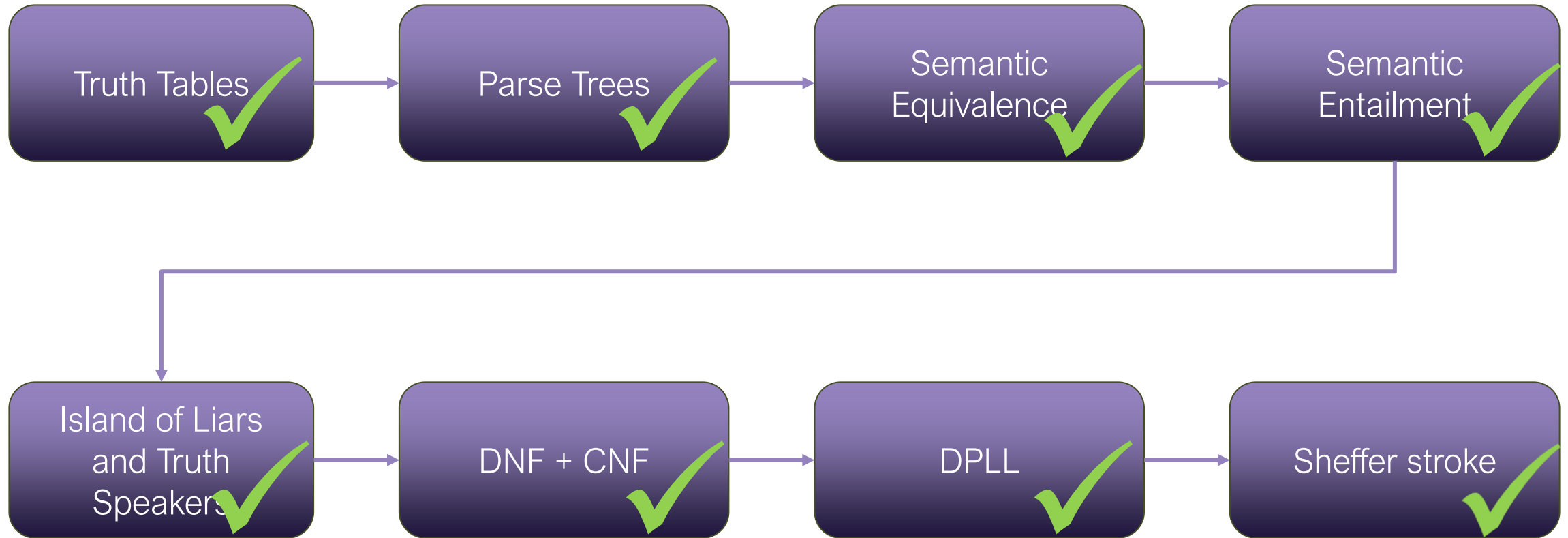
Counterexample:

If $\phi = T, \psi = T, \chi = F$

Then $(\phi|\psi)|\chi$ is true, but $\phi|(\psi|\chi)$ is false

So, the Sheffer stroke is **not** associative

Planning Day 1



Planning for tomorrow

Sunday 	Logic week 1 + 2 + 3
Tuesday	Sets week 1: ○ Set operations ○ Venn diagrams Sets week 2: ○ Algebra of sets ○ Relations Sets week 3: ○ Ordering relations ○ Hasse Diagrams ○ Cartesian and Lexicographic Order

Questions?

Join the WhatsApp group for slides! Invite link should be in your mail, or use the QR code

If you have any questions, ask them in the WhatsApp group :)

See you all again on Tuesday!

WhatsApp

